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LOAD FREQUENCY STABILIZATION USING OPTIMIZED MPC-(1+PIDN) CONTROLLER FOR HIGH ORDER INTERCONNECTED RENEWABLE ENERGY BASED POWER SYSTEMS

Abstract

A load frequency control system for a multi-area hybrid power grid having multiple regions which integrates renewable energy sources. The system includes individual power systems in separate geographic regions, interconnected by electrical tie-lines. Area control signals (ACE) generated by each power system facilitate adaptive frequency stabilization. A central model predictive controller (MPC) processes ACE signals to produce control signals, guiding a one plus proportional integral derivative (1+PIDN) controller with adjustable gain parameters. An optimizer applies a salp swarm algorithm to the ACE signals to dynamically adjust gain parameter of the 1+PIDN controller. The resulting output signal from the 1+PIDN controller is designed to mitigate power disturbances and maintain frequency balance across the tie-lines, ensuring robust and responsive grid management.

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Background/Summary

STATEMENT REGARDING PRIOR DISCLOSURE BY THE INVENTORS

[0001] Aspects of this technology are described in an article “Designing Of Robust Frequency Stabilization Using Optimized MPC-(1+PIDN) Controller For High Order Interconnected Renewable Energy Based Power Systems” published in Protection and Control of Modern Power Systems 8, in January 2023, which is incorporated herein by reference in its entirety.

CROSS REFERENCE TO RELATED APPLICATIONS

[0002] The present application claims the benefit of priority to U.S. Prov. App. No. 63/562,529, entitled “Designing Of Robust Frequency Stabilization Using Optimized MPC-(1+PIDN) Controller For High Order Interconnected Renewable Energy Based Power Systems”, filed on Mar. 7, 2024, and incorporated herein by reference in its entirety.

STATEMENT OF ACKNOWLEDGEMENT

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BACKGROUND

Technical Field

[0004] The present disclosure is directed to a controller for high-order interconnected renewable energy based power system.

Description of Related Art

[0005] The “background” description provided herein is for the purpose of generally presenting the context of the disclosure. Work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description which may not otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present invention.

[0006] Adoption of renewable energy sources is on the rise owing to their potential to offer a practically limitless supply of clean energy, aligning with global efforts to combat climate change and reduce carbon emissions. However, the variable power output of these resources presents significant challenges. Solar energy production fluctuates due to diurnal cycles and weather conditions, while wind energy generation fluctuates due to varying wind speeds and directions. This intermittency creates a mismatch between renewable energy generation and peak consumer demand periods, complicating the integration of renewable energy sources into existing power grid infrastructures.

[0007] Frequency stabilization in renewable energy system (RES) networks is essential for maintaining grid integrity and ensuring a reliable power supply. Electrical frequency, measured by the number of alternating current cycles per second in the grid, must adhere to strict limits based on regional standards. The need for frequency stabilization arises from inherent characteristics of RES. Unlike traditional power plants with predictable outputs, renewable energy systems such as solar and wind, are inherently intermittent, leading to rapid and unpredictable fluctuations in power generation. These fluctuations can cause imbalances between power supply and demand, resulting in frequency variations. Such deviations from the nominal frequency can disrupt power generation, transmission, and distribution efficiency, potentially causing damage to infrastructure and end-user equipment.

[0008] In RES-dependent power grids, the challenge of frequency stabilization is heightened due to the stochastic nature of renewable energy production. For instance, solar panels produce less energy during cloudy periods, and wind turbine output is contingent on wind conditions. This dependence can cause sudden drops or spikes in power generation, which may destabilize the power system if not managed effectively,

[0009] To address destabilization challenges, frequency stabilization strategies often combine fast-acting energy storage systems with advanced control mechanisms such as load frequency control (LFC) and automatic generation control (AGC). These controllers regulate generator outputs and coordinate network resources to maintain frequency within the required range. Frequency fluctuations stem from load-generation imbalances and nonlinear factors such as generation rate constraints (GRC). LFC and AGC manage both frequency and power interchange across tie-lines in interconnected regions to balance loads and power generation.

[0010] Extensive research has focused on controller performance in complex energy networks. Literature highlights various controllers for LFC, with PI/PID controllers commonly deployed for frequency stabilization. PI/PID controllers play a crucial role in maintaining nominal frequency in power systems, impacting operational stability significantly. Optimizing PI/PID controllers aims to enhance their performance in complex power network environments. However, the response time of these optimized controllers may still be suboptimal in intricate networks characterized by diverse energy inputs and demand fluctuations.

[0011] Various metaheuristic techniques have been employed to derive optimal parameters for these controllers, using algorithms, such as the bacterial foraging optimization algorithm (BFOA), the firefly algorithm (FA), and genetic algorithms (GA) in the AGC controller design. Furthermore, alternative control structures have been investigated to enhance effectiveness. In an example, integral double derivative (IDD) controllers, when operated by a lightning search algorithm (LSA), have been applied for LFC, though additional oscillations are introduced in the

frequency response. Controllers based on integral order (IO), intelligence, and degrees of freedom, including 2DOF and the advanced 2DOF, have also been introduced to augment AGC performance.

[0012] In scenarios of multi-area power systems, which are inherently more complex due to the interaction between multiple energy-producing regions, cascaded controllers have been implemented. The cascaded controllers have been proven effective in demonstrating their capability to maintain system stability. However, cascaded controllers can be complex to design and tune. The process of adjusting multiple control parameters to achieve desired system behavior is intricate and often requires sophisticated computational techniques. Optimization algorithms, such as the grey wolf optimizer (GWO) and the bat algorithm (BA), have been applied to tune the PI-PD controllers, exhibiting their ability to fine-tune control parameters in pursuit of optimal system behavior, including the mitigation of frequency oscillations in electric vehicles. The optimization algorithms search through a wide solution space to find optimal settings. Such algorithms can be computationally challenging, particularly for real-time applications.

[0013] High-complexity algorithms, such as the Harris Hawks optimization have shown advancements, but present a high level of complexity that can be a barrier to practical implementation.

[0014] Additionally, hybrid algorithms combining stochastic fractal search and pattern search techniques (hSFS-PS) have also been implemented to advance the control capability of PI-PD controllers. Nevertheless, many of these techniques were tested within simplified systems, predominantly limited to two-area networks.

[0015] To address the deficiencies of existing controllers, fuzzy/adaptive based PI/PID controller tuning has been used. However, the integration of fuzzy controllers introduces complexity. Internal model controllers (IMC) based on PI design have been developed to address LFC challenges, and advanced methods, such as dropout deep neural networks with transfer learning have been utilized for optimal AGC dispatch. Moreover, dispatch techniques using a model predictive controller (MPC) framework have been disclosed for maximizing total profit by dynamic variation in the AGC. The MPC is especially valued for its rapid response and robustness against load disruption and uncertainty. The application of MPCs has been explored in thermal systems and for systems connected to photovoltaic (PV) installations, yet the performance in complex networks has not been fully analyzed. Recent studies have introduced MPC-PI gain scheduling and adaptive AGC dispatch methods for renewable energy integration to enhance frequency regulation in more complex multi-area systems. Active disturbance rejection control (LADRC) has also been designed for such systems to address LFC issues effectively.

[0016] Thus, it is identified that conventional control techniques, such as proportional-integral-derivative (PI/PID) controllers, face challenges in diminishing uncertainty due to system errors. Cascaded formation designs have been introduced to improve controller performance, yet do not robustly address this uncertainty. The complexity of power system networks, particularly in higher-order areas, results in significant gaps in the analysis and efficacy of controller designs for such systems.

[0017] US20130268131A1 describes a multi-area power control system that is designed to manage power distribution across interconnected regions. The system monitors various parameters such as frequency, tie-line power, and total active power loss, which are essential for maintaining grid stability. However, this reference does not incorporate a self-adaptive learning particle (SALP) algorithm, nor does it include the MPC or the 1+PID controller design.

[0018] CN112636368B describes an automatic power generation control method for a multi-source, multi-area interconnected power system. The method describes a fractional order PID controller that is optimized using the celestial beard algorithm. Although this reference provides a sophisticated approach to power generation control, this reference lacks the integration of a cascaded MPC with a 1-PIDN controller and does not utilize SALP optimization.

[0019] US20050127680A1 describes a central controller system tasked with monitoring power generation across multiple areas. It emphasizes the importance of centralized monitoring and management for stable power distribution. However, this system does not discuss the innovative approach of a cascaded MPC with a 1-PIDN controller, nor does it mention the implementation of a SALP optimization technique.

[0020] Each of the aforementioned patents presents advancements in power system control but also possesses limitations in their scope and capability, failing to address the specific innovations of the present invention. The identified patents do not suggest a comprehensive solution that combines the cascaded MPC with a 1-PIDN controller design, along with SALP optimization, to effectively manage the dynamic and complex nature of multi-area interconnected power systems.

[0021] Thus, there exists a need for an integrated approach to enhance the control and stability of power systems that integrate multiple energy sources and regions. The present invention aims to fulfill this need by providing a novel control system that not only encompasses the necessary advancements over the prior art but also introduces a methodical and simplified design process that obviates the complexities associated with current power control technologies. Consequently, it is an object of the present disclosure to offer a method and system for the integration of multi-area power systems with renewable energy sources, which employs a model predictive controller cascaded with a 1-PIDN controller for improved performance and efficiency.

SUMMARY

[0022] In an exemplary embodiment, a load frequency control system for a multi-region power system incorporating a plurality of renewable energy sources is described. The load frequency control system includes a first power system located in a first geographic region, a second power system located in a second geographic region, a third power system located in a third geographic region, a fourth power system located in a fourth geographic region, a first electrical tie-line configured to connect the first power system with the second power system, a second electrical tie-line configured to connect the first power system with the third power system, a third electrical tie-line configured to connect the second power system with the third power system, and a fourth electrical tie-line configured to connect the first power system with the fourth power system. Each power system is configured to generate area control signals (ACE). The load frequency control system further includes a model predictive controller (MPC) operatively connected to receive the ACE signals from each electrical tie-line. The MPC includes a memory storing program instructions, and at least one processor configured to execute the program instructions to generate a control signal $u(k)$. The load frequency control system further includes a one plus proportional integral derivative (1+PIDN) controller operatively connected to the MPC. The 1+PIDN controller includes a set of amplifiers having gain parameters. The load frequency control system further includes an optimizer operatively connected to the MPC. The optimizer is configured to receive the ACE signals, apply the ACE signals to a salp swarm algorithm and update the gain parameters of the 1+PIDN controller. The 1+PIDN controller is configured to receive the control signal $u(k)$ and the updated gain parameters and generate an output signal to mitigate power disturbances due to frequency imbalances on the first electrical tie line, the second electrical tie-line, the third electrical tie-line, and the fourth electrical tie-line.

[0023] In another exemplary embodiment, a method for load frequency control of a multi-region power system incorporating a plurality of renewable energy sources is described. The method includes connecting a first power system located in a first geographic region to a second power system located in a second geographic region by a first electrical tie-line, connecting the first power system to a third power system located in a third geographic region by a second electrical tie-line, connecting the second power system to the third power system by a third electrical tie-line, and connecting the first power system to a fourth power system located in a fourth geographic region by a fourth electrical tie-line. The method further includes generating, by each power system, area control signals (ACE) and receiving, by a model predictive controller (MPC), the ACE signals from each power system. The MPC includes a memory storing program instructions and at least one processor configured to execute the program instructions. The method further includes generating, by the MPC, a control signal $u(k)$, connecting a one plus proportional integral derivative (1+PIDN) controller, which includes a set of amplifiers having gain parameters to the MPC, and connecting an optimizer to the MPC. The method further includes receiving, by the optimizer, the ACE signals, applying, by the optimizer, the ACE signals to a salp swarm algorithm, and updating, by the optimizer, the gain parameters. The method further includes receiving, by the 1+PIDN controller, the control signal $u(k)$ and the updated gain parameters and generating, by the 1+PIDN controller, an output signal to mitigate power disturbances due to frequency imbalances on the first electrical tie line, the second electrical tie-line, the third electrical tie-line, and the fourth electrical tie-line.

[0024] In another exemplary embodiment, a hybrid load frequency controller for interconnected multi-region power systems incorporating a plurality of renewable energy sources is described. The hybrid load frequency controller includes a model predictive controller (MPC) operatively connected to receive area control error (ACE) signals from electrical tie-lines which interconnect the multi-region power systems. The MPC includes a memory and at least one processor configured to execute the program instructions to generate a control signal $u(k)$. The hybrid load frequency controller further includes one plus proportional integral derivative (1+PIDN) controller operatively connected to the MPC. The 1+PIDN controller includes a set of amplifiers having gain parameters. The hybrid load frequency controller further includes an optimizer operatively connected to the MPC. The optimizer is configured to receive the ACE signals, apply the ACE signals to a salp swarm algorithm and update the gain parameters of the 1+PIDN controller. The 1+PIDN controller is configured to receive the control signal $u(k)$ and the updated gain parameters and generate an output signal configured to mitigate power disturbances due to frequency imbalances on the electrical tie-lines which interconnect the multi-region power systems.

[0025] The foregoing general description of the illustrative embodiments and the following detailed description thereof are merely exemplary aspects of the teachings of this disclosure, and are not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0026] A more complete appreciation of this disclosure and many of the attendant advantages thereof will be readily obtained as the same becomes better understood by reference to the following detailed description when considered in connection with the accompanying drawings, wherein:

[0027] FIG. 1 illustrates a block diagram of a load frequency control (LFC) system implemented within a multi-area

hybrid power model, according to certain embodiments.

[0028] FIG. 2 illustrates a pole-zero map of a control system for a fourth power system corresponding to a fourth geographic area, according to certain embodiments.

[0029] FIG. 3 illustrates a block diagram of a photovoltaic (PV) system interconnected to the LFC system for a multi-region power system, according to certain embodiments.

[0030] FIG. 4 illustrates a block diagram of a wind energy unit, according to certain embodiments.

[0031] FIG. 5A illustrates a schematic diagram of a controller integrating a model predictive controller (MPC) with a 1+proportional integral derivative plus number (1+PIDN) controller, according to certain embodiments.

[0032] FIG. 5B illustrates a configuration of a load frequency control system designed for four interconnected power systems, according to certain embodiments.

[0033] FIG. 6 illustrates a flowchart for implementing a salp swarm algorithm (SSA) used in an MPC combined with a 1+PIDN controller in a Simulink model, according to certain embodiments.

[0034] FIG. 7 illustrates a graphical representation of abrupt load deviations over time, according to certain embodiments.

[0035] FIG. 8A illustrates the frequency response of the controller implemented in a first geographic area, according to certain embodiments.

[0036] FIG. 8B illustrates an exploded view of a section of the frequency response of the controller of FIG. 8A, according to certain embodiments.

[0037] FIG. 9A illustrates the frequency response of the controller implemented in a second geographic area, according to certain embodiments.

[0038] FIG. 9B illustrates an exploded view of a section of the frequency response of the controller of FIG. 9A, according to certain embodiments.

[0039] FIG. 10A illustrates the frequency response of the controller implemented the a third geographic area, according to certain embodiments.

[0040] FIG. 10B illustrates an exploded view of a section of the frequency response of the controller of FIG. 10A, according to certain embodiments.

[0041] FIG. 11A illustrates the frequency response of the controller implemented in a fourth geographic area, according to certain embodiments.

[0042] FIG. 11B illustrates an exploded view of a section of the frequency response of the controller of FIG. 11A, according to certain embodiments.

[0043] FIG. 12 depicts a frequency deviation graph comparing the performance of various controllers within a power system, according to certain embodiments.

[0044] FIG. 13 illustrates a tie-line power-sharing response in a first power system of a first geographic area for various controllers, according to certain embodiments.

[0045] FIG. 14 presents a tie-line power-sharing response of various controllers implemented within a second power system, according to certain embodiments.

[0046] FIG. 15 illustrates a tie-line power-sharing response of the various controllers implemented within a third power system, according to certain embodiments.

[0047] FIG. 16 illustrates a tie-line power-sharing response of the various controllers implemented within a fourth power system, according to certain embodiments.

[0048] FIG. 17 illustrates a load deviation response of a controller over time for a system subjected to random load variations, according to certain embodiments.

[0049] FIG. 18 illustrates a comparison of the MPC-(1+PIDN) controller with various controllers of the frequency response in the first power system implemented in the first geographic area, according to certain embodiments.

[0050] FIG. 19 illustrates a comparison of the MPC-(1+PIDN) controller with various controllers of the frequency response of the second power system implemented in the second geographic area, according to certain embodiments.

[0051] FIG. 20 illustrates a comparison of the MPC-(1+PIDN) controller with various controllers of the frequency response of the third power system implemented in the third geographic area, according to certain embodiments.

[0052] FIG. 21 illustrates a comparison of the MPC-(1+PIDN) controller with various controllers of the frequency response of the fourth power system implemented in the fourth geographic area, according to certain embodiments.

[0053] FIG. 22 illustrates a comparison of the MPC-(1+PIDN) controller with various controllers of a tie-line response between the first geographic area and the second geographic area, according to certain embodiments.

[0054] FIG. 23 illustrates a comparison of the MPC-(1+PIDN) controller with various controllers of a tie-line response between the first geographic area and the third geographic area, according to certain embodiments.

[0055] FIG. 24 illustrates a comparison of the MPC-(1+PIDN) controller with various controllers of a tie-line response between the second geographic area and the third geographic area, according to certain embodiments.

[0056] FIG. 25 a comparison of the MPC-(1+PIDN) controller with various controllers of illustrates a tie-line response between the first geographic area and the fourth geographic area, according to certain embodiments.

[0057] FIG. 26A illustrates the PV power output versus time, according to certain embodiments.

[0058] FIG. **26B** illustrates the deviation in wind power output versus time, according to certain embodiments.
[0059] FIG. **26C** illustrates the deviation in hydroelectric plant power output versus time, according to certain embodiments.

[0060] FIG. **26D** illustrates the deviation in the combined hydroelectric plant plus PV plus wind power output versus time, according to certain embodiments.

[0061] FIG. **27** illustrates details of computing hardware of the cascaded MPC 1+PIDN controller, according to certain embodiments.

[0062] FIG. **28** is an exemplary schematic diagram of a data processing system used within the computing system, according to certain embodiments.

[0063] FIG. **29** is an exemplary schematic diagram of a processor used with the computing system, according to certain embodiments.

[0064] FIG. **30** is an illustration of a non-limiting example of distributed components which may share processing with the controller, according to certain embodiments.

DETAILED DESCRIPTION

[0065] In the drawings, like reference numerals designate identical or corresponding parts throughout the several views. Further, as used herein, the words “a”, “an” and the like generally carry a meaning of “one or more”, unless stated otherwise.

[0066] Furthermore, the terms “approximately,” “approximate”, “about” and similar terms generally refer to ranges that include the identified value within a margin of 20%, 10%, or preferably 5%, and any values therebetween.

[0067] A geographic area, alternately referred to as a geographic region, is defined as an portion of the surface of the earth which is bounded by latitudinal and longitudinal coordinates. Each geographic area may have a plurality of electrical power generators which provide power to a load center located in the geographic area. The power generators of each geographic area are operatively connected to a common utility grid. Tie-lines, as described below, interconnect the power systems of the different geographic areas. In the present disclosure, the geographic area may be alternately be referred to as a geographic region, a geographic power area or a geographic power region.

[0068] In the present disclosure, the plurality of electrical power generators may include thermal power generators as well as renewable energy generators, which include photovoltaic (PV) arrays, wind turbines and hydroelectric power generators.

[0069] Thermal power generation consists of using steam power created by burning oil, liquid natural gas (LNG), coal, and other substances to rotate generators and create electricity. Thermal power generation plays a central role in supplying power because it can flexibly respond to the various ways in which power is used (demand fluctuations) as the power output grows larger. Aspects of this disclosure are directed to a load frequency control (LFC) system and a method for load frequency control of a power network. The LFC system includes four interconnected power systems located in different geographic areas, each with an assortment of power generation sources, including photovoltaic (PV), wind, thermal, and hydroelectric power generators. The stability of the LFC system is dependent on the effectiveness of the LFC controller, especially with the varying input from renewable sources and the challenges posed by the inclusion of a hydroelectric power system, which introduces specific dynamics into the power system which require a robust control solution.

[0070] To address these challenges, the LFC controller is embodied as a model predictive controller (MPC) cascaded with a one plus proportional integral derivative with filter (1+PIDN) controller. A salp swarm algorithm (SSA) is implemented for the optimization of the parameters of the 1+PIDN controller. The MPC-(1+PIDN) controller is specifically configured to stabilize frequency within high-order interconnected areas. The MPC-(1+PIDN) controller results in rapid frequency stabilization across all four areas, faster than traditional control methods.

[0071] FIG. **1** illustrates a block diagram of a load frequency control (LFC) system **100** implemented within a multi-area hybrid power network. The LFC system **100** is integrated within a grid-connected, multi-area power infrastructure designed to optimally distribute and stabilize power frequency across various distinct geographic areas. The hybrid model is one illustration of power management systems designed to integrate various combinations of energy resources, including renewable energy resources and traditional energy resources, to manage the effective distribution and stabilization of power frequency across various geographic zones.

[0072] The LFC system **100** can be implemented in a plurality of geographic areas. The geographic area may be a city, a district, a county, a combination of towns, and the like. In one illustration, FIG. **1** demonstrates the LFC system **100** as applied to a four-area power network, incorporating multiple energy resources integrated to serve the energy requirements of each geographic area. The LFC system **100** includes a first power system **102** located in a first geographic area (area-1), a second power system **104** located in a second geographic area (area-2), a third power system **106** situated in a third geographic area (area-3), and a fourth power system **108** positioned in a fourth geographic area (area-4). The first, second, and third power systems **102**, **104**, and **106** are integrated with a combination of renewable energy resources, such as wind, solar, and thermal energy resources, and the fourth power system **108** is integrated with wind, solar, and hydroelectric power energy resources.

[0073] Each power system **102**, **104**, **106** and **108** includes respective energy-generating units and associated loads.

Specifically, the first power system **102** includes at least three energy resources, including a first thermal generator **118** and a first renewable energy unit **120** which integrates a photovoltaic energy generator and a wind energy generator. The first thermal generator **118** is connected to a traditional energy generator side busbar **102-1**. The first photovoltaic energy generator and a first wind energy generator are connected to a renewable energy generator side busbar **102-2**.

[0074] The generator side busbar is the point where the electrical power generated by the various generating units, such as thermal, wind, and hydro, within a power system area, is collected. The generator side busbar is a common electrical junction to which the outputs of all generators are connected. The voltage level at the generator side busbar is typically high because it serves as a collection point for the generated power before it is transmitted or stepped up for transmission over long distances.

[0075] From the traditional energy generator side busbar **102-1**, the power generated by the thermal unit **118** is collected by a load busbar **102-9**. Similarly, from the renewable energy generator side busbar **102-2**, the power generated by the first renewable energy unit **120** is collected by a load busbar **102-10**.

[0076] The load busbar is the point of distribution where the power is delivered from the power system to the different loads. The power at the load busbar has usually been stepped down to a lower voltage suitable for local distribution. It distributes power to various consumers, including any of residential, commercial and industrial users. In FIG. 1, the power collected by the load busbar **102-9** and the load busbar **102-10** is delivered to an area-1 **102** busbar **102-11**. The busbar **102-11** is connected to a load **122**. The busbar **102-11** is also referred to as the first busbar. The load **122** represents the energy consumers in the power system **102** of area-1. The busbar **102-11** (first busbar) is connected to one or more of the electrical tie-lines (for example, tie-line **12**, tie-line **13**, tie-line **14** and tie-line **23**). The power disturbances in the tie-lines are due to changes in a demand of the load and fluctuation in the energy generating resources located in the power system of area-1 **102**.

[0077] The busbar **102-11** is further connected to receive the power disturbances from each of the first electrical tie-line **12**, the second electrical tie-line **13**, and the fourth electrical tie-line **14**. As shown in FIG. 5B, the first power system **102** also includes a frequency meter **512** configured to measure a frequency on each of the first electrical tie-line **12**, the second electrical tie-line **13** and the fourth electrical tie-line **14** and generate a combined frequency measurement. The first power system **102** further includes a first subtractor **514** configured to subtract the combined frequency measurement from a reference frequency **516** and generate a power to frequency ratio bias signal **524**. The first power system **102** further includes an interchange scheduler **518** configured to generate a time window for an exchange of power from the first busbar to the MPC-(1+PIDN) controller **508**. Further, the first power system **102** includes a second subtractor **520** configured to receive the power disturbances from the first busbar and transmit the power disturbances during the time window. The first power system **102** further includes a third subtractor **522** configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate an ACE.sub.1 signal.

[0078] Referring back to FIG. 1, the second power system **104** of area-2 incorporates at least three energy resources, including a second thermal generator **124**, a second renewable energy unit **126** that integrates a photovoltaic energy generator and a wind energy generator. The second thermal generator **124** is connected to an energy generator side busbar **104-3**. The second photovoltaic energy generator and a second wind energy generator are connected to a renewable energy generator side busbar **104-4**.

[0079] Power generated by the thermal generator **124** is collected at a load busbar **104-12**, while power from the second renewable energy unit **126** is gathered at a load busbar **104-13**. The load busbars **104-12** and **104-13** deliver power to a busbar **104-14**, which is then connected to load **128**. The busbar **104-14** is also referred to as the second busbar or the second busbar **104-14**. The load **128** represents various energy consumers within the second geographic region.

[0080] In one aspect, the second busbar **104-14** is connected to receive the power disturbances from each of the first electrical tie-line **12** and the third electrical tie-line **23**. Referring to FIG. 5B, the second power system **104** also includes a frequency meter **528** configured to measure a frequency on each of the first electrical tie-line **12** and the third electrical tie-line **23** and generate a combined frequency measurement at adder **526**. In FIG. 5B, the first electrical tie-line **12** and the third electrical tie-line **23** are shown as tie-lines **21** and **23** respectively, reflective of the direction of power flow. The second power system **104** further includes a first subtractor **530** configured to subtract the combined frequency measurement from a reference frequency **530** and generate a power to frequency ratio bias signal **540**. The second power system **104** further includes an interchange scheduler **534** configured to generate a time window for an exchange of power from the second busbar **104-14** to the MPC-(1+PIDN) controller **508**, which is connected to each of the power systems. Further, the second power system **104** includes a second subtractor **536** configured to receive the power disturbances from the second busbar and transmit the power disturbances at the scheduled time. The second power system **104** further includes a third subtractor **538** configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate an ACE.sub.2 signal which is sent to the MPC-(1+PIDN) controller **508**.

[0081] The third power system **106** of area-3 includes a combination of energy resources, including a third thermal

generator **130**, and a third renewable energy unit **132** integrating a photovoltaic energy generator and a wind energy generator. The third thermal generator **130** is linked to a traditional energy generator side busbar **106-5**. The third photovoltaic energy generator and the third wind energy generator are connected to a renewable energy generator side busbar **106-6**.

[0082] The traditional energy generator side busbar **106-5** channels the electricity produced by the third thermal unit **130** to a load busbar **106-15**, while the renewable energy generator side busbar **106-6** channels electricity from the third renewable energy unit **132** to a load busbar **106-16**. These load busbars **106-15** and **106-16** are responsible for delivering power to a third busbar **106-17**, which supplies energy to load **134**, servicing various consumers in the third geographic region. The busbar **106-17** is also referred to as the third busbar or the third busbar **106-17**.

[0083] In one aspect, the third busbar **106-17** is connected to receive the power disturbances from each of the first electrical tie-line **12** and the third electrical tie-line **23**. As shown in FIG. 5B, the third power system **106** also includes a frequency meter **548** configured to measure a frequency on each of the second electrical tie-line **13** and the third electrical tie-line **23** and generate a combined frequency measurement. In FIG. 5B, the second electrical tie-line **13** and the third electrical tie-line **23** are illustrated as tie-lines **31** and **32** respectively, reflective of the direction of power flow. The third power system **106** further includes a first subtractor **530** configured to subtract the combined frequency measurement from a reference frequency **532** and generate a power to frequency ratio bias signal **556**. The third power system **106** further includes an interchange scheduler **550** configured to generate a time window for an exchange of power from the third busbar **106-17** to the MPC-(1+PIDN) controller **508**. Further, the third power system **106** includes a second subtractor **536** configured to receive the power disturbances from the third busbar and transmit the power disturbances at the scheduled time. The third power system **106** further includes a third subtractor **538** configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate an ACE.sub.3 signal.

[0084] The fourth power system **108** of area-4 is integrated with a set of energy resources, including a hydroelectric power generator **136** and a fourth renewable energy unit **138** including a photovoltaic energy generator and a wind energy generator. The hydroelectric power generator **136** is connected to a generator side busbar **108-7**. The fourth photovoltaic energy generator and the fourth wind energy generator are connected to a renewable energy generator side busbar **108-8**.

[0085] Electricity from the hydro generator **136** is directed to a load busbar **108-18**, while the renewable energy units direct the power output to a load busbar **108-19**. Power from these load busbars **108-18** and **108-19** is consolidated at a fourth busbar **108-20**, which then feeds a load **140**. The load **140** includes a variety of energy consumers located in the fourth geographic region. The busbar **108-20** is also referred to as the fourth busbar or the fourth busbar **108-20**.

[0086] In one aspect, the fourth busbar **108-20** is connected to receive the power disturbances due to the fluctuation in the generation of power by the renewable energy generators **138** within area-4, fluctuations in demand from load **140** and from the fourth electrical tie-line **14**. Referring to FIG. 5B, the fourth power system **108** includes a frequency meter **560** configured to measure a frequency on the fourth electrical tie-line **14** and generate a frequency measurement. In FIG. 5B, the fourth electrical tie-line **14** is illustrated as tie-line **41** respectively, reflective of the direction of power flow. The fourth power system **108** further includes a first subtractor **562** configured to subtract the combined frequency measurement from a reference frequency **564** and generate a power to frequency ratio bias signal **572**. The fourth power system **108** further includes an interchange scheduler **566** configured to generate a time window for an exchange of power from the fourth busbar **108-20** to the MPC-(1+PIDN) controller **508**. Further, the fourth power system **108** includes a second subtractor **568** configured to receive the power disturbances from the fourth busbar and transmit the power disturbances at the scheduled time. The fourth power system **108** further includes a third subtractor **570** configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate an ACE.sub.4 signal.

[0087] Connectivity between the power systems is achieved through the network of electrical tie-lines. An electrical tie-line in power systems is a transmission line that connects two different power grids or systems. By such connection, power between the systems connected by the tie-line is exchanged in order to balance supply and demand across wider areas. The first electrical tie-line **12** is configured to connect the first power system **102** with the second power system **104**, the second electrical tie-line **13** connects the first power system **102** with the third power system **106**, a third electrical tie-line **23** interlinks the second power system **104** with the third power system **106**, and a fourth electrical tie-line **14** connects the first power system **102** with the fourth power system **108**.

[0088] Each power system **102**, **104**, **106**, and **108** is further configured to produce and transmit the area control error (ACE) signals. ACE.sub.1 is the ACE signal generated by the first power system, ACE.sub.2 is the ACE signal generated by the second power system, ACE.sub.3 is the ACE signal generated by the third power system, and ACE.sub.4 is the ACE signal generated by the fourth power system. The ACE is a quantity used in operating bulk electric systems. The ACE is defined as the instantaneous difference between a balancing net actual and scheduled interchange with all adjacent interconnected balancing authority areas. The ACE is measured in megawatts (MW). The ACE signals, resultant from the aggregation of power deviations, offer a metric of discrepancy from the

predetermined frequency set-point. The ACE signal is fed to the MPC-(1+PIDN) controller **508** of the LFC system **100** to analyze the deviations in supply and demand in the energy distribution network, and accordingly, to balance the power distribution in the grid and compensate for fluctuations in the power signals.

[0089] The ACE signals are configured as inputs for the MPC-(1+PIDN) controller **508** of the LFC system **100**. As shown in FIG. 5A, the MPC is the first stage of the controller. The MPC is equipped with memory and a processor having processing capabilities to execute program instructions for the generation of a control signal $u(k)$. The MPC is a controller based on logic that uses a mathematical model of a system to predict future behaviour and optimize control actions. At each time step, the MPC forecasts future outcomes based on current states and control inputs, then calculates the control inputs to achieve the desired performance according to a predefined objective function. In one example, the sampling time of the MPC is 0.1 seconds, the prediction horizon is 10 and the control horizon is 3.

[0090] The output of MPC, control signal $u(k)$, is fed to a second stage of the controller. The second stage is configured with a one-plus proportional integral derivative (1+PIDN) controller. The 1+PIDN controller is operatively interconnected with the MPC, configured as a series of amplifiers with tunable gain parameters, which are dynamically updated by an optimizer. The 1+PIDN controller is configured to receive the control signal $u(k)$ and the updated gain parameters and generate an output signal configured to mitigate power disturbances due to frequency imbalances on the first electrical tie line **12**, the second electrical tie-line **23**, the third electrical tie-line **13**, and the fourth electrical tie-line **14**.

[0091] The 1+PIDN controller is a type of controller that builds upon the proportional-integral-derivative (PID) control strategy. The “1+” indicates an additional component in the control structure, typically representing a filter configured for improving the robustness or performance of the controller.

[0092] The PIDN controller is a PID controller that includes a derivative filter, denoted by “N” to improve the derivative action. The derivative filter is used to mitigate the effects of noise in the derivative term, which can lead to erratic control signals. The PIDN controller thus enhances the ability of the classic PID controller to predict and compensate for future trends in the behavior of the system, which is particularly useful in systems with higher-order dynamics or where the derivative of the error signal is noisy.

[0093] When combined with an MPC, the 1+PIDN controller may be used to fine-tune the control actions based on the output of the predictive model, thereby providing both the predictive capabilities of the MPC and the reactive, compensatory actions of the PID controller with enhanced filtering through the derivative term. Thus, the controller, a combination of the MPC and the 1+PIDN controllers, results in a control system that can effectively anticipate system needs while maintaining stability and reacting to immediate changes in the behavior of the LFC system **100**.

[0094] In one aspect, the LFC system **100** further includes an optimizer **504** operatively connected to the MPC. The optimizer is configured to receive the ACE signals, apply the ACE signals to a salp swarm algorithm which updates the gain parameters of the 1+PIDN controller **506**. The controller, in one aspect, is configured to optimize the system by the salp swarm algorithm implemented to refine the gain parameters based on the received ACE signals.

[0095] The salp swarm algorithm is a computational algorithm inspired by the swarming behavior of salps in the ocean. Salps move in a swarm in a chain-like formation, a behavior that is translated into an algorithmic context to solve optimization problems. With reference to the present disclosure, the salp swarm algorithm is implemented to optimize the parameters of the 1+PIDN controller for adjusting power distribution across the power grid. In an example, if the frequency of the power grid starts to fall because demand is outstripping supply, the grid sensors would detect this and could send signals to increase the output of certain generators. Conversely, if the frequency rises above the nominal value due to an excess of supply over demand, the grid sensors would instruct some generators to scale back their output. The MPC-1+PIDN controller **508** in the LFC system **100** could be used to fine-tune these adjustments by predicting future system states and adjusting control inputs accordingly to ensure smooth, stable operation without oscillations or overshoots based on the salp swarm algorithm.

[0096] In the LFC system **100**, as depicted in FIG. 1, each energy generation unit has a specific configuration of parameters for energy generation. Moreover, the grid also accounts for nonlinearities. Most of these nonlinearities are composed of a rate limiter and a generator rate constraint (GRC). Each energy unit can be described based on configurational parameters and nonlinearities computation. Each energy unit is described in the subsequent description.

[0097] The thermal unit includes components, such as the electrical generator, a governor, a steam turbine, and a reheater. The governor attached to the generator is configured to maintaining frequency stability during load imbalances. In one implementation of the present disclosure, in a model of the thermal system, an example of a combined power plant capacity of 2200 MW for each area, with a standard operational load of 2000 MW and one plant running at 1000 MW is used. The base power level is set at 1000 MVA.

[0098] The governor output $\Delta P_{\text{sub.gi}}(s)$ is the difference of the reference power $\Delta P_{\text{sub.ref}}$ and frequency change $\Delta f_{\text{sub.i}}(s)$ passed through droop control, $1/R$ is given as:

$$[00001] \quad P_{\text{gi}}(s) = P_{\text{ref}}(s) - \frac{1}{R} f_i(s); \quad (1)$$

[0099] Power transfer between the areas (i and j) can be summarized as:

$$[00002] \quad P_{\text{tie},ij} = \text{Math.}_{j=1}^n \frac{T_{ij}}{S} (f_i - f_j); i \neq j \quad (2)$$

[0100] Area central error (ACE) for each area is given by:

$$[00003] \quad \text{ACE}_i = B_i f_i + P_{\text{tie},ij}; i \neq j \quad (3) \quad [0101] \text{ where } B_{\text{sub},i} \text{ is the frequency bias factor parameter in p.u. of MW/Hz, and } \Delta P_{\text{sub},\text{tie},ij} \text{ is the change of tie-line power between two areas } i \text{ and } j.$$

[0102] The output $\Delta f_{\text{sub},i}$ is expressed as:

$$[00004] \quad f_i(s) = G_p(s) [\text{Math.}_{j=1}^n P_{Rij} - P_{d,i} - P_{\text{tie},i}]; \text{ where } G_p(s) = \frac{1}{M_i s + D_i} \quad (4)$$

[0103] The fourth area comprises a hydroelectric power plant along with PV and wind energy resources. The hydroelectric power plant contributes to meeting the energy demand of the overall power generating system, while the PV and wind also contribute to meeting the energy demand. The nominal operation load on the hydroelectric power plant is 1000 MW, while the installed capacity of the hydro power plant is 2000 M W. The hydroelectric power plant exchanges power with the first geographic area-1 that is further connected with the second geographic area-2 and the third geographic area-3.

[0104] The hydro turbine transfer function is expressed as:

$$[00005] \quad G_t(s) = \frac{K_r T_r s + 1}{T_r s + 1} \cdot \text{Math.} \frac{-T_w s + 1}{0.5 T_w s + 1} \quad (5)$$

[0105] For the fourth geographic area-4, the open loop transfer function can be written as:

$$[00006] \quad G(s) = \frac{K_{p4}(T_2 s + 1)(-T_w s + 1)}{(T_1 s + 1)(T_3 s + 1)(T_4 s + 1)(0.5 T_w s + 1)} \quad (6)$$

[0106] The model symbols along with their values are listed in Table 1.

TABLE-US-00001 TABLE 1 Model symbols Terminology Symbols Values Frequency bias factor B.sub.1, B.sub.2, B.sub.3, B.sub.4 0.425 MW/Hz Speed regulation constant R.sub.1, R.sub.2, R.sub.3, R.sub.4 2.4 Hz/MW Generator gain K.sub.p1, K.sub.p2, K.sub.p3, K.sub.p4 120 80 Generator time constant T.sub.p1, T.sub.p2, T.sub.p3, T.sub.p4 20 s 13 s Governor time constant T.sub.g1, T.sub.g2, T.sub.g3, T.sub.1 0.2 s 48.7 s Hydro turbine transmission T.sub.2, T.sub.3 0.513 s coefficient 10 s Initial time constant for T.sub.w 1 s water flow Reheat time constant T.sub.r 20 s Reheat gain K.sub.r 0.333

[0107] FIG. 2 illustrates a pole-zero map of the MPC one plus proportional integral derivative (1+PIDN) controller **508** implemented for the fourth power system corresponding to the fourth geographic area. The graphical representation elucidates the control system dynamics by displaying the system poles, denoted by 'X', and zeros, denoted by 'O'.

[0108] In the pole-zero map, the zero **202** is located on the right half of the plane, indicating a non-minimum phase element within the system. The non-minimum phase element is an indicative of instability. Correspondingly, the pole **204**, situated on the left half of the plane, typically contributes to stability of the system.

[0109] The LFC system **100**, including the MPC-(1+PIDN) controller **508**, is adapted to address the stabilization challenges posed by the system dynamics identified in FIG. 2. The MPC, which receives area control error (ACE) signals, and the 1+PIDN controller, with gain parameters updated by an optimizer which applies a salp swarm algorithm, function collaboratively to produce control signals capable of compensating for power imbalances.

[0110] This configuration ensures the robust control necessary for the stability of the entire four-area system, inclusive of the renewable and traditional energy sources within the power system. The pole-zero map is considered in the design and implementation to provide the requisite information for configuration of the LFC system **100** to maintain the stability of the power across all interconnected areas.

[0111] FIG. 3 illustrates a block diagram of a photovoltaic (PV) system interconnected to the load frequency control LFC system **100** for a multi-region power system.

[0112] The photovoltaic system comprises a photovoltaic panel **302**, which serves as a direct current (DC) nonlinear output source. The photovoltaic panel **302** may represent an array of photovoltaic panels, for example, a field of photovoltaic panels. The performance of the photovoltaic panel **302** is contingent on solar irradiation, which can vary due to atmospheric conditions, as illustrated by the presence of both the sun and clouds. This variability impacts the design and operation of the PV system within the power system, particularly considering the need to integrate with other energy sources in a stable and efficient manner.

[0113] To facilitate the integration of the PV generation into the power system, the PV model includes several conversion units. The initial stage of conversion is handled by a DC-DC boost converter **304**, which receives the direct current (I.sub.1) from the photovoltaic panel **302**. The DC-DC boost converter **304** is configured for stepping up the voltage level to match the requirements of the subsequent power conversion stages.

[0114] Following this, the DC power (I.sub.2) is conveyed to a current inverter DC-AC **306**, which converts the direct current into alternating current (I.sub.AC). This conversion is essential for integrating the DC power produced by the photovoltaic panel **302** into the predominantly AC power grid.

[0115] The alternating current is then processed through a computation model **308** for conversion of instantaneous power, which is implemented to translate the alternating current into an immediate measure of power (P).

[0116] The PV system includes a subsystem **310** for obtaining the average power, which calculates the average

power (P.sub.avg) output to be fed into the grid. This average power reflects the overall performance of the PV system over time, smoothing out the fluctuations inherent in instantaneous power measurements due to the varying input from solar irradiation.

[0117] Each component of the PV system is configured to ensure that the photovoltaic system operates at maximum efficiency, transforming the variable DC power into stable AC power suitable for grid integration. The design and configuration of the PV system, as described, are vital for achieving the objectives of the load frequency control system as claimed, providing a reliable and consistent energy source within the larger multi-region power system.

[0118] In the PV model of FIG. 3, the PV is considered to be operating at its maximum power point. The transfer function of the boost converter is given as:

[00007] $g_1(s) = \frac{1}{m_1}$ (7) [0119] where m.sub.1 is:

$$[00008] m_1 = \frac{v_2}{v_1} = \frac{i_1}{i_2} \quad (8)$$

[0120] Similarly, the inverter transfer function is denoted as:

$$[00009] g_2(s) = \frac{I_{ac}(s)}{I_2(s)} = \frac{s^2}{s^2 + \omega^2} \quad (9) \quad [0121] \text{ where } \omega = 2\pi f = 2\pi(50) = 314.12 \text{ rad/(sec.)}$$

[0122] The transfer function of the instantaneous power is given as:

$$[00010] P(s) = \frac{v_m i_m}{2s} + \frac{v_m i_m}{2} \frac{s}{s^2 + (2\omega)^2} \quad (10) \quad [0123] \text{ where, m represents the mean, i.e., the mean voltage v.sub.m and the mean current i.sub.m.}$$

[0124] The gain of the instantaneous power is provided as:

$$[00011] g_3(s) = \frac{p(s)}{I_{ac}(s)} = v_m \left(\frac{(s^2 + \omega^2)(s^2 + (2\omega)^2)}{s^2(s^2 + (4\omega)^2)} \right) \quad (11)$$

[0125] The average power is expressed as:

$$[00012] p_{avg}(s) = \frac{v_m i_m}{2s} \quad (12) \quad [0126] \text{ and the average power gain is described as:}$$

$$[00013] g_4(s) = \frac{p_{avg}(s)}{p(s)} \quad (13)$$

[0127] FIG. 4 illustrates a schematic representation of the wind energy unit, in accordance with the present disclosure. The wind energy conversion model is designed to address the complexities of integrating wind energy into a multi-region power system.

[0128] The model begins with the wind velocity V_w, an input to the aerodynamic model block 402, where β represents the blade pitch angle and λ the tip speed ratio. The aerodynamic model, defined by equation (14), computes the mechanical power P_m based on these inputs, which characterizes the energy captured from the wind.

[0129] The mechanical output power of the wind turbine (WT), including the wake effect is expressed as:

$$V_{wake} = V_m \left[1 - \prod_{i=1}^n \left(1 - \sqrt{1 - \frac{F_T}{r^2 V_m^2}} \right) \left(\frac{r}{r+kx} \right)^2 \right]$$

$$[00014] P_w = \frac{1}{2} A_T C_p(\beta, \lambda) V_{wake}^3 \quad (14) \quad \text{where, } \left\{ \begin{array}{l} C_p(\beta, \lambda) = C_1 \left(\frac{C_2}{C_1} - C_3 - C_4 \right) e^{-\frac{C_5}{\lambda}} + C_6 - T \\ \frac{1}{T} = \frac{1}{+0.08} - \frac{0.035}{\lambda^3 + 1}; \quad T = \frac{w_r r}{V_w}; \quad A_T = \frac{d^2}{4} \end{array} \right. \quad [0130]$$

where C.sub.p is the power coefficient, β is the blade pitch angle, and ρ denotes the air density (Kg/m³). The rotor diameter is represented by d (m), and the tip speed ratio (TSR) is denoted by λ.sub.T and λ.sub.I is the intermittent TSR. F.sub.T is a thrust force, k denotes the decreasing coefficient (for onshore WT, k=0.075, and for offshore WT, k=0.05), Δx is the distance between two consecutive WTs, and V.sub.w is the effective wind velocity.

From the aerodynamic model block 402, the mechanical power P_m flows into the two-mass model of the shaft block 404. This model, governed by equation (15), captures the dynamic behavior of the wind turbine shaft, considering the mass and elasticity of the shaft system that transfers the mechanical energy from the turbine blades to the generator.

[0131] The two-mass model of the gear coupling is represented as:

$$\begin{aligned} \dot{\theta}_{WT} &= -\left(\frac{D_{WT} + D_{shaft}}{J_{WT}} \right) \theta_{WT} + \left(\frac{D_{shaft}}{J_{WT}} \right) \theta_{gen} - \left(\frac{1}{J_{WT}} \right) T_{int} + \left(\frac{1}{J_{WT}} \right) d(t) \\ [00015] \left\{ \begin{array}{l} \dot{\theta}_{gen} = \left(\frac{D_{shaft}}{J_{WT}} \right) \theta_{WT} + \left(\frac{D_{gen} + D_{shaft}}{J_{gen}} \right) \theta_{gen} - \left(\frac{1}{n_{gear} J_{gen}} \right) T_{int} + \left(\frac{1}{J_{gen}} \right) u(t) \\ T_{int} = K_{shaft} \left(\theta_{WT} - \frac{\theta_{gen}}{n_{gear}} \right) \end{array} \right. \quad (15) \quad [0132] \text{ where } J_{sub.WT} \text{ and } J_{sub.gen} \text{ represent WT moment} \end{aligned}$$

of inertia of the rotor and generator, respectively. Shaft internal torque is represented by T.sub.int, WT rotor and

generator speed is denoted by $\omega_{\text{sub.r,WT}}$, and gear turns ratio is represented by $n_{\text{sub.gear}}$. $D_{\text{sub.shaft}}$ and $D_{\text{sub.gen}}$ are the damping constant of the WT, coupling shaft and generator, respectively.

[0133] The mechanical power P_m output from the two-mass shaft model block **404** is then input into the generator block **406**. This block is defined by equations (16-17) and represents the conversion of mechanical energy into electrical power P_w , which is the final output of the wind energy conversion process and is fed into the power grid.

[0134] Assuming the output power $P_{\text{sub.w}}$ is ideal and the generator is lossless, the equations are:

[00016] $P_w = T_{\text{gen}} \omega_g$; (16) $\frac{dT_{\text{gen}}}{dt} = (-\frac{1}{\tau_{\text{gen}}})T_{\text{gen}} + (\frac{1}{\tau_{\text{gen}}})T_{\text{control}}$; (17) [0135] where $\tau_{\text{sub.gen}}$ denotes the time constant of the generator, and $T_{\text{sub.control}}$ represents the control torque that manipulates the output power of the WT against uncertain wind speed.

[0136] The precise modeling of each stage of the wind energy conversion model, as illustrated, ensures that the wind energy contribution to the power system is accurately controlled to mitigate the effects of wind variability on the frequency stability of the power system.

[0137] FIG. 5A illustrates a schematic diagram of the LFC system **500** which integrates a MPC **502** with a 1+proportional integral derivative number (1+PIDN) controller **506**, in accordance with the present disclosure. The LFC system **500** is implemented to manage the stability and distribution of power within a multi-region power system.

[0138] The MPC **502**, as described earlier with reference to FIGS. 1 and 5B, is a controller that receives the ACE of each area as an input, in addition to a predefined reference signal. With the inputs, the MPC **502** is configured to execute a cost-minimization strategy that respects operational constraints. The MPC controller **502** is implemented for forecasting the system behavior and adjusting control efforts proactively to maintain the desired power system equilibrium.

[0139] The MPC controller **502** includes three blocks. First, a predictive block; second, an optimization solver; and third, a cost function block. The predictive model captures the relationship between the system's inputs, outputs, and internal states. It forecasts how the system will respond to different control actions and disturbances. The predictive block utilizes a dynamic model of the system to predict its future behavior over a defined time horizon. Based on the future behavior of the system, the MPC controller **502** formulates an optimization problem to determine the optimal control actions that minimize or maximize a certain objective function over the prediction horizon while satisfying system constraints. The optimization solver finds the control inputs that achieve a best trade-off between performance and constraint satisfaction. The cost function block defines an objective that the MPC controller **502** seeks to optimize. The cost function block quantifies a desired performance criteria, such as minimizing energy consumption, maximizing production, or tracking a reference trajectory. The cost function includes terms that penalize deviations from setpoints, control effort, and violation of constraints.

[0140] The cost function is applied to the SSA to be minimized.

[0141] The cost function $J_{\text{sub.ITAE}}$ is an integral time absolute error given by:

[00017]

$$J_{\text{ITAE}} = \int_0^{T_{\text{sin}}} t (\text{Math. ACE}_1(t) \cdot \text{Math.} + \text{Math. ACE}_2(t) \cdot \text{Math.} + \text{Math. ACE}_3(t) \cdot \text{Math.} + \text{Math. ACE}_4(t) \cdot \text{Math.}) dt$$

[0142] where t is time, $\text{ACE}_{\text{sub.1}}$ is the ACE signal generated by the first power system, $\text{ACE}_{\text{sub.2}}$ is the ACE signal generated by the second power system, $\text{ACE}_{\text{sub.3}}$ is the ACE signal generated by the third power system, $\text{ACE}_{\text{sub.4}}$ is the ACE signal generated by the fourth power system. ITAE is defined as the integral time-weighted absolute error.

[0143] The function of these blocks is to shift the system response towards the reference trajectory based on the past and future predictions.

[0144] The output predictive vector is represented by $Y_{\text{sub.p}}(k)$ as:

$$Y_p(k) = Z(k) + U(k) + \sum_I U_I(k) \quad (18)$$

[0145] The control law is obtained by the gradient descent method, i.e., $(\partial J(k))/\Delta U_{\text{sub.}}(k)=0$. To obtain the control law $u(k)$ the following equations are applied:

$$U(k) = (Q^T + R)^{-1} Q^T (Y_r(k) - Z(k) - \sum_I U_I(k)) \quad (19) \quad u(k) = (E_{N_u} O_{N_u(P-1)})^T U(k) \quad (20)$$

$$u(k) = u(k) + u(k-1) \quad (21)$$

[0146] In the design of the cost function for the LFC problem, the constraint for the MPC is expressed as:

$$\min J(k) = \min \{ (Y_p(k) - Y_r(k))^T Q (Y_p(k) - Y_r(k)) + (u(k))^T R (u(k)) \} \quad (22)$$

[0147] The parameters Q and R are the weighting vectors to balance the square future control and performance predictive error.

[0148] The optimization of the control parameters is conducted by the salp swarm algorithm **504**. The algorithm adjusts the control gains parameters, including proportional gain (K_p), integral gain (K_i), derivative gain (K_d), and a filter coefficient (N). The gain parameters define the responsiveness and stability of the 1+PIDN controller **506**.

[0149] The control signal $u(k)$ that is the output of the MPC is fed to (1+PIDN), and the (1+PIDN) controller processes the optimal signal of the MPC to further refine the signal to achieve the optimal control output for the LFC calculation, given by:

[00021] $\text{output} = (1 + K_P + \frac{K_I}{s} + \frac{K_D N s}{N + s}) \times u(k)$ (23) [0150] where $u(k)$ is the output signal of the MPC.

[0151] The 1+PIDN controller **506** integrates the gain parameters into its control logic. The proportional gain (K_p) modulates response of the controller proportionally to the error signal, offering immediate corrective action. The integral gain ($K_{\text{sub.I}}$) addresses the accumulation of past errors, resulting in the elimination of residual steady-state errors. The derivative gain (K_d) provides a predictive control action based on the rate of change of the system error, enhancing the ability of the system to anticipate future errors. Finally, the filter coefficient (N) is applied to the derivative term to mitigate the impact of high-frequency noise on the control signal, refining the control action for a smoother system response.

[0152] Together, the components of the 1+PIDN controller **506** operate in concert to expedite the control process, effectively reducing settling time of the system and adeptly managing frequency disruptions. The controller design, detailed in FIGS. 5A, 5B, emphasizes not only the individual contributions of each control parameter but also their collective impact on performance of the system, ensuring rapid and robust load frequency control in a complex, interconnected power system environment.

[0153] FIG. 5B illustrates the configuration of a load frequency control system designed for four interconnected power systems, each situated within a separate geographic region.

[0154] In the first power system **508-1**, the busbar **510** is the point of convergence for power disturbances from tie-lines **12**, **13**, and **14**. A frequency meter **512** is configured for measuring frequency across these tie-lines and creating an aggregate frequency measurement. A first subtractor **514** receives the aggregate frequency measurement and compares the aggregate frequency measurement to a frequency reference signal **516** to generate a power to frequency ratio bias signal **524**. An interchange scheduler **518** is configured to establish the timing for a power transfer from the busbar **510** to the MPC-(1+PIDN) controller **508**. Following this, a second subtractor **520** processes the disturbances from the busbar **510**, scheduling the disturbances based on the plan of the interchange scheduler **518**. A third subtractor **522** receives the power to frequency ratio bias signal **524** and subtracts the scheduled power disturbances to produce the automatic generation control signal (AGC).

[0155] In the second power system **508-2**, the busbar **526** is the point of convergence for power disturbances from tie-lines **21** and **23**. A frequency meter **528** is configured for measuring frequency across these tie-lines and creating an aggregate frequency measurement. A first subtractor **530** receives the aggregate frequency measurement and compares the aggregate frequency measurement to a frequency reference signal **532** to generate a power to frequency ratio bias signal **540**. An interchange scheduler **534** is configured to establish the timing for a power transfer from the busbar **526** to the MPC-(1+PIDN) controller **508**. Following this, a second subtractor **536** processes the disturbances from the busbar **526**, scheduling the disturbances based on the plan of the interchange scheduler **534**. A third subtractor **522** receives the power to frequency ratio bias signal **540** and applies it to the scheduled power disturbances to produce the AGC.

[0156] In the third power system **508-3**, the busbar **542** is the point of convergence for power disturbances from tie-lines **31** and **32**. A frequency meter **544** is configured for measuring frequency across these tie-lines and creating an aggregate frequency measurement. A first subtractor **546** receives the aggregate frequency measurement and compares the aggregate frequency measurement to a frequency reference signal **548** to generate a power to frequency ratio bias signal **556**. An interchange scheduler **550** is configured to establish the timing for a power transfer from the busbar **542** to the MPC-(1+PIDN) controller **500**. Following this, a second subtractor **552** processes the disturbances from the busbar **542**, scheduling the disturbances based on the plan of the interchange scheduler **550**. A third subtractor **554** receives the power to frequency ratio bias signal **556** and applies it to the scheduled power disturbances to produce the AGC.

[0157] In the fourth power system **508-4**, the busbar **558** is the point of convergence for power disturbances from a tie-line **41**. A frequency meter **560** is configured for measuring frequency across these tie-lines and creating an aggregate frequency measurement. A first subtractor **562** receives the aggregate frequency measurement and compares the aggregate frequency measurement to a frequency reference signal **564** to generate a power to frequency ratio bias signal **572**. An interchange scheduler **566** is configured to establish the timing for a power transfer from the busbar **558** to the MPC-(1+PIDN) controller **508**. Following this, a second subtractor **568** processes the disturbances from the busbar **558**, scheduling the disturbances based on the plan of the interchange scheduler **566**. A third subtractor **570** receives the power to frequency ratio bias signal **572** and applies to the scheduled power disturbances to produce the AGC.

[0158] Each of these power systems integrates an MPC-(1+PIDN) configuration, demonstrating an advanced system devised to ensure the efficient management of frequency and power distribution across an interconnected network.

[0159] FIG. 6 illustrates a flowchart of a method for implementing the salp swarm algorithm (SSA) with a MPC combined with a 1+Proportional Integral Derivative plus Number (1+PIDN). In a non-limiting example, the method

is described as simulated with a Simulink model in MATLAB 2017a, using a CoreTMi7 CPU at 2.2 GHz and 8 GB of RAM. The SSA implementation contributes to the robust load frequency control in a multi-region power system. [0160] At step **602**, the sequence is initiated. Next, step **604** integrates the Simulink model with SSA MATLAB code, establishing a computational foundation for the optimization process. At step **606**, the SSA controlling parameters are specified, which will guide the search process within the algorithm, by programming the Simulink model with the cost function J.sub.ITAE and entering the parameters for the salps, such as velocity and initial position. At step **608**, the initial population is generated randomly, providing a diverse set of potential solutions from which the algorithm can commence the optimization. At step **610**, the fitness function for each salp is evaluated, and the algorithm determines the salp according to the defined fitness criteria, which the cost function J.sub.ITAE based on the area control error (ACE) values of each area within the power system. Step **612** involves updating the value of coefficient C1 using the predefined equation (25), an action that adjusts the search behavior of the algorithm in the solution space. The iterative process of the algorithm is handled in steps **614** through **630**. Within this loop, the positions of the leading and following salps are updated in steps **618** and **620**, respectively, utilizing equations (24) and (26) to refine their locations in the search space based on the fitness evaluations. The position of each salp is recorded at step **622**, providing a historical record of progression of the algorithm. The iterative loop continues with a check at step **624** to determine whether all salps have been evaluated in the current iteration. Upon reaching the maximum number of iterations, as evaluated at step **628**, the optimized model parameters are displayed at step **632**, signifying the conclusion of optimization phase of the algorithm. The model is then simulated with the obtained optimal parameters at step **634**, ensuring that the system operates according to the optimized settings. Finally, step **636** compares the results with well-known methods in the literature for validation to evaluate the performance of the SSA-optimized MPC-(1+PIDN) controller. Such comparison serves to substantiate the efficacy of the controller design in managing the load frequency control problem within the interconnected power system.

[0161] The salp swarm optimization algorithm is designed from the transparent bodied salp which transforms itself into a chain-like structure during the search for food. The salp swarm algorithm is mainly comprised of two sections: (i) leader and (ii) followers. The salp swarm algorithm is translated in the form of a block diagram shown in FIG. 6. [0162] The leader movement of the SSA is decided by:

$$[00022] K_j^i = \begin{cases} M_i + C_1((ub_j - lb_j)C_2 + lb_j), & C_3 \geq 0 \\ M_i - C_1((ub_j - lb_j)C_2 + lb_j), & C_3 < 0 \end{cases}; \quad (24) \quad [0163] \text{ where } M \text{ and } K \text{ signify the target food and}$$

two-dimensional salp position, respectively. $ub_{sub.j}$ and lb_j denote the respective upper and lower bounds of the $j_{sup.th}$ dimension. Similarly, $C_{sub.2}$ and $C_{sub.3}$ are uniform coefficient numbers and $C_{sub.1}$ is further expressed as:

$$[00023] C_1 = 2e^{-(\frac{4t}{t_{max}})^2}; \quad (25) \quad [0164] \text{ where } t \text{ denotes the current iteration and } t_{sub.max} \text{ signifies the maximum number of iterations. The position is updated by:}$$

$$[00024] K_j^i = \frac{1}{2}at^2 + v_0 t; s \quad (26) \quad [0165] \text{ where, } K \text{ represents gain parameters.}$$

[0166] FIG. 1 to FIG. 6 illustrate the LFC system for a multi-region power system incorporating a plurality of renewable energy sources. The LFC system was simulated in MATLAB 2017a, using a CoreTMi7 CPU at 2.2 GHz and 8 GB of RAM. The simulation results are depicted in FIG. 7 through FIG. 26.

[0167] In the simulation environment, the various algorithms of GA, GWO, SSA, and MPC were initialized, and the initial algorithm parameters configuration details are stated in Table 2.

TABLE-US-00002 TABLE 2 Algorithms parameters setting
Algorithm Parameters
GA Population size = 50, maximum generations = 100, crossover probability = 0.75, mutation probability = 0.1. GWO Wolf number = 50, maximum iteration = 100, $\alpha = 0.7$, $\gamma = 0.9$ SSA Population size = 50, maximum iteration = 100, $C_2 = 0.5$, $C_3 = 0.5$ MPC Prediction horizon $P = 3$, control horizon $M = 2$,

[0168] Further, the optimized controller parameters for the four-area based power system are formulated in Table 3.

TABLE-US-00003 TABLE 3 Optimized controller parameters
Controller MPC- Parameters
GA-PI GWO:PI-PD
MPC/PI (1 + PIDN)
Area-1 $K_{sub.P} = -0.6130$ $K_{sub.P} = -2.823$ $K_{sub.P} = 0.12$ $K_{sub.P} = 4.89$ $K_{sub.I} = -0.8010$ $K_{sub.I} = -2.789$ $K_{sub.I} = 0.81$ $K_{sub.I} = 7.710$ $K_{sub.P} = -3.084$ $K_{sub.D} = 0.0148$ $K_{sub.D} = -1.997$ $N = 300$
Area-2 $K_{sub.P} = -0.6911$ $K_{sub.P} = -2.902$ $K_{sub.P} = 0.17$ $K_{sub.P} = 7.310$ $K_{sub.I} = -0.7665$ $K_{sub.I} = -3.627$ $K_{sub.I} = 0.92$ $K_{sub.I} = 12.927$ $K_{sub.P} = -3.084$ $K_{sub.D} = 0.102$ $K_{sub.D} = -0.982$ $N = 300$
Area-3 $K_{sub.P} = -0.6025$ $K_{sub.P} = -3.1240$ $K_{sub.P} = 0.20$ $K_{sub.P} = 2.267$ $K_{sub.I} = -0.7901$ $K_{sub.I} = -3.1121$ $K_{sub.I} = 0.32$ $K_{sub.I} = 6.126$ $K_{sub.P} = -3.0401$ $K_{sub.D} = 0.0241$ $K_{sub.D} = -0.8012$ $N = 300$
Area-4 $K_{sub.P} = -0.6611$ $K_{sub.P} = -2.8114$ $K_{sub.P} = 0.11$ $K_{sub.P} = 6.589$ $K_{sub.I} = -0.8021$ $K_{sub.I} = -3.6101$ $K_{sub.I} = 0.41$ $K_{sub.I} = 12.106$ $K_{sub.P} = -3.1144$ $K_{sub.D} = 0.0015$ $K_{sub.D} = -1.1212$ $N = 300$

[0169] FIG. 7 illustrates a graphical representation of abrupt load deviations over time. The abrupt deviations are calculated to assess the performance of the control technique under high-order areas (HOA) within a power system.

[0170] FIG. 7 plots load deviation (p.u.) on the vertical axis against time (seconds) on the horizontal axis. Notably

marked by reference number **702**, the plotted step changes indicate instances of load perturbation introduced to test the robustness of the control system. These deviations range from 1% to 5%, with the upper limit representing a severe condition in load variation. The depicted step changes, line **702**, simulates the dynamics of the power system and the response of the controller to such fluctuations.

[0171] FIG. **8A** illustrates the frequency response of the controller configured for the first geographic area within a multi-area power system. The frequency response determines the comparison of the performance of various controllers, including the MPC-(1+PIDN) controller. In the graph, the vertical axis represents the frequency deviation in per unit (p.u.) and the horizontal axis denotes the time in seconds.

[0172] FIG. **8A** displays several response curves, each corresponding to different controller configurations. A curve **802** corresponds to a genetic algorithm-proportional integral (GA-PI) controller. A curve **804** is the response of a grey wolf optimizer-proportional integral derivative (GWO-PID) controller. A curve **806** corresponds to a firefly algorithm-proportional integral Derivative (FA-PID) controller. A curve **808** denotes the response of the MPC-(1+PIDN) controller.

[0173] The dashed section is an area of particular interest, and is enlarged in FIG. **8B** for detailed analysis. Within this region, transient behaviors of the control systems under load variations are magnified, providing insight into the comparative effectiveness of the controllers under test conditions.

[0174] The MPC-(1+PIDN) controller demonstrates enhanced stability and a rapid return to the nominal frequency value following a disturbance, which is characterized by its curve marked with reference number **808**. Such performance is indicative of the of the controller and ability to maintain system equilibrium under varying load conditions.

[0175] FIG. **9A** presents the frequency response of the controller implemented for the second geographic area. As with FIG. **8A**, the response curves of various controllers are plotted to demonstrate their efficacy in stabilizing frequency deviations.

[0176] A curve **902** depicts the response of the GA-PI controller. A curve **904** represents the GWO-PID controller. A curve **906** corresponds to the FA-PID controller. A curve **908** depicts the response of the MPC-(1+PIDN) controller, highlighting its performance.

[0177] The dashed section is an area of particular interest, which is enlarged in the FIG. **9B**. The lower section is a detailed view of the transient response phase, where effectiveness the control strategies in mitigating frequency excursions is evaluated. Response curve **908** of the MPC-(1+PIDN) controller illustrates its ability to dampen oscillations and quickly restore system frequency to its desired value following a load change.

[0178] FIG. **10A** illustrates the frequency response of the controller implemented for the third geographic area, examining the load frequency control capabilities of the MPC-(1+PIDN) controller in comparison to other algorithms.

[0179] The graph in FIG. **10A** includes response curves for the various control strategies tested. Curve **1002** corresponds to the GA-PI controller. Curve **1004** signifies the GWO-PID controller. Curve **1006** represents the FA-PID controller. The response of the MPC-(1+PIDN) controller is shown with curve **1008**.

[0180] Within the graph, the dashed section is an area of particular interest, which focuses on transient performance, where the control methodologies are subjected to abrupt load changes. The response curve this area is enlarged and presented in FIG. **10B**. The response **1008**, pertaining to the MPC-(1+PIDN) controller, shows a reduction in overshoot and a swift stabilization period, evidencing efficiency of the controller.

[0181] FIG. **11A** depicts the frequency response of the MPC-(1+PIDN) controller for the fourth geographic area, especially in managing the complexities introduced by the right half-plane zero in the system.

[0182] Response curves for different control strategies are plotted against time. Curve **1112** shows the response of the GA-PI controller. Curve **1114** illustrates the GWO-PID controller response. Curve **1116** represents the FA-PID controller response. Response of the MPC-(1+PIDN) controller is depicted by curve **1118**.

[0183] The dashed section is an area of particular interest as is the response curve that reflects behaviour of the system in reaction to significant load perturbations. The response curve for the dashed area is enlarged in FIG. **11B**. The MPC-(1+PIDN) controller response, is indicated by curve **1118** that demonstrates robust control capability of the controller, particularly in achieving rapid stabilization of frequency post-disturbance.

[0184] The results of the simulations are compiled to analyze the effectiveness of the MPC-(1+PIDN) controller and to compare with other state-of-the-art controllers as listed in Table 4.

TABLE-US-00004 TABLE 4 Performance analysis of the different controllers MPC-(1 + PIDN) GWO:PI-PD MPC/PI GA-PI Controller S.T U.S O.S S.T U.S O.S S.T U.S O.S S.T U.S O.S Area-1 0.86 -0.001 0 3.3 -0.03 0.01 9.57 -0.03 0 17.87 -0.18 0.01 Area-2 1.08 0 0.01 2.93 0 0.05 8.61 0 0.04 20.1 -0.01 0.18 Area-3 0.81 -0.01 0 4.45 -0.04 0.01 9.76 -0.02 0 17.93 -0.18 0.01 Area-4 0.84 0 0.02 3.26 -0.02 0.06 13.5 -0.006 0.05 18 -0.12 0.24

[0185] In Table 4, S.T represents the response time, U.S represents the undershoot amplitude and O.S represents the overshoot amplitude.

[0186] FIG. **12** depicts a frequency deviation graph comparing the performance of various controllers within a power system. The vertical axis measures frequency deviation in per unit (p.u.), and the horizontal axis represents

time in seconds.

[0187] FIG. **12** includes four distinct data plots. The first plot **1212**, illustrates the frequency response of the Genetic Algorithm-Proportional Integral (GA-PI) controller, showing significant high frequency deviation. The second plot **1214**, shows the response for the Grey Wolf Optimizer-Proportional Integral Derivative (GWO-PI-PD) controller with frequency deviation at lower magnitudes. The third plot **1218**, represents the frequency response of the Model Predictive Control-Proportional Integral (MPC-PI) controller. The fourth plot **1216**, demonstrates the frequency response for the MPC-(1+PIDN) controller, exhibiting minimal frequency deviation, which approaches zero.

[0188] FIG. **13** illustrates the tie-line **12** power-sharing response of the various controllers implemented within the first power system, comparing the MPC-(1+PIDN) controller of the disclosure against other conventional techniques.

[0189] FIG. **13** includes four line plots representing different control strategies, each annotated with a unique reference number for identification. A curve **1312** corresponds to the GA-PI controller, a curve **1314** to the GWO-PI-PD controller, a curve **1318** to the MPC-PI controller, and a curve **1316** illustrates the response of the MPC-(1+PIDN) controller. The curves demonstrate how each controller manages power-sharing in response to load changes, with the MPC-(1+PIDN) controller (plot **1316**) showcasing effective power-sharing as evidenced by the smaller amplitude of fluctuations.

[0190] FIG. **14** presents a graphical analysis of the tie-line **23** power-sharing response of the various controllers implemented within the second power system, where each plot is associated with a specific control strategy.

[0191] FIG. **14** displays a response of the GA-PI controller illustrated as a curve **1412**, a response of GWO-PI-PD controller as a curve **1414**, response of the MPC-PI controller as a curve **1418**, and a response of the MPC-(1+PIDN) controller as plot **1416**. The MPC-(1+PIDN) controller maintains power-sharing close to the nominal range, indicating effective frequency stabilization.

[0192] FIG. **15** illustrates the tie-line **13** power-sharing response of the various controllers implemented within the third power system. The frequency deviations are depicted for the GA-PI controller, by curve **1512**, the GWO-PI-PD controller, by curve plot **1514**, the MPC-PI controller, by a curve **1518**, and the MPC-(1+PIDN) controller, by a curve **1516**. The MPC-(1+PIDN) controller shows a more stable response with fewer oscillations and quicker stabilization, emphasizing its robustness.

[0193] FIG. **16** illustrates the tie-line **14** power-sharing response of the various controllers implemented within the fourth power system. In this illustration, the responses are depicted for the GA-PI controller, by a curve **1612**, the GWO-PI-PD controller, by a curve **1614**, the MPC-PI controller, by a curve **1618**, and the MPC-(1+PIDN) controller, by a curve **1616**. The response curve **1616** of the MPC-(1+PIDN) controller, reflects a proficient power-sharing capability, with quick damping of fluctuations and minimal overshoot and undershoot.

[0194] The response time (ST) of the MPC-(1+PIDN) controller is 0.86 s and has an undershoot response (US) of -0.001 for the first geographic area. In comparison, the response of GWO: PI-PD controller is 3.3 s while also exhibiting US and an overshoot (OS) of -0.03 and 0.01 respectively in the first geographic area. Similarly, the response of MPC/PI presents a settling time (ST) of 9.57 seconds and a US response of -0.03 with no OS. For the first geographic area, the response manifested by GA-PI is 17.87 seconds to settle the frequency fluctuation, while US and OS are -0.18 and 0.01 , respectively. In the case of the second geographic area, the controller shows an ST of 1.08 s and OS of 0.01 with no US. In comparison, the ST, US and OS of GWO: PI-Pd are 2.93 seconds, 0 and 0.05 , respectively. In the case of the MPC/PI controller, the ST is 9.57 seconds, US is -0.03 with no OS. The GA-PI controller shows ST of 20.1 seconds and the responses of US and OS are -0.01 and 0.18 , respectively. The simulated response of the MPC-(1+PIDN) in the third geographic area shows an ST of 0.81 seconds, US of -0.01 with no OS. Similarly, in third geographic area, the ST responses of GWO: PI-PD, MPC/PI and GA-PI are 4.45 seconds, 9.76 seconds and 17.93 seconds, respectively. The US of the GWO: PI-PD, MPC/PI and GA-PI are -0.04 , -0.02 and -0.18 , respectively. Further, the OS response of GWO: PI-PD is 0.01 , while the MPC/PI presents no OS and GA-PI exhibits 0.01 OS response.

[0195] The responses of the MPC-(1+PIDN), GWO: PI-PD, MPC/PI and GA-PI in the fourth geographic area show STs of 0.84 seconds, 3.26 seconds, 13.5 seconds and 18 seconds, while the US are 0, -0.02 , -0.006 and -0.12 , respectively. In case of the OS responses, the MPC-(1+PIDN), GWO: PI-PD, MPC/PI and GA-PI depict 0.02 , 0.06 , 0.05 , and 0.24 , respectively.

[0196] FIG. **17** illustrates the load deviation response of a controller over time for a system subjected to random load variations. Curve **1702** indicates a response of the MPC-(1+PIDN) controller **508** to a random changing load between 1% to 6% variations.

[0197] FIG. **18** illustrates the frequency response of the first power system implemented for the first geographic area. Various curves compare the efficacy of several control strategies over time. Curve **1802** depicts the GA-PI controller response, curve **1804** shows the response of the GWO: PI-PD controller, and curve **1806** outlines the MPC-PI controller response. The MPC-(1+PIDN) controller response is presented by curve **1808**. Curve **1808** demonstrates the robustness of the MPC-(1+PIDN) controller that suppresses the frequency fluctuation in minimum time compared to other controllers.

[0198] FIG. **19** illustrates the frequency response of the second power system implemented for the second geographic area. Various curves compare the efficacy of several control strategies over time. Curve **1902** depicts the GA-PI controller response, curve **1904** shows the response of the GWO: PI-PD controller, and curve **1906** illustrates the MPC-PI controller response. The MPC-(1+PIDN) controller response is presented by curve **1908**. Curve **1908** demonstrates the robustness of the MPC-(1+PIDN) controller that suppresses the frequency fluctuation in minimum time compared to other controllers.

[0199] FIG. **20** illustrates the frequency response of the third power system implemented for the third geographic area. Various curves compare the efficacy of several control strategies over time. Curve **2002** depicts the GA-PI controller response, curve **2004** shows the response of the GWO: PI-PD controller, and curve **2006** illustrates the MPC-PI controller response. The MPC-(1+PIDN) controller response is presented by curve **2008**. Curve **2008** demonstrates the robustness of the MPC-(1+PIDN) controller that suppresses the frequency fluctuation in minimum time compared to other controllers.

[0200] FIG. **21** illustrates the frequency response of the fourth power system implemented for the fourth geographic area. Various curves compare the efficacy of several control strategies over time. Curve **2102** depicts the GA-PI controller response, curve **2104** shows the response of the GWO: PI-PD controller, and curve **2106** illustrates the MPC-PI controller response. The MPC-(1+PIDN) controller response is presented by curve **2108**. Curve **2108** demonstrates the robustness of the MPC-(1+PIDN) controller that suppresses the frequency fluctuation in minimum time compared to other controllers.

[0201] FIG. **22** illustrates an electrical tie-line **12** response between the first geographic area and the second geographic area. Connectivity between the first power system and the second power system is achieved through the electrical tie-line **12**. Various curves shows response by various controllers to sharing of power between two geographic areas through the tie-line. Curve **2202** depicts the GA-PI controller response, curve **2204** shows the response of the GWO: PI-PD controller, and curve **2206** illustrates the MPC-PI controller response. The MPC-(1+PIDN) controller response is presented by curve **2208**. Curve **2208** demonstrates the most efficient sharing of power between the corresponding geographic areas.

[0202] FIG. **23** illustrates an electrical tie-line **13** response between the first geographic area and the third geographic area. Connectivity between the first power system and the third power system is achieved through the electrical tie-line **13**. Various curves shows response by various controllers to sharing of power between two geographic areas through the tie-line. Curve **2302** depicts the GA-PI controller response, curve **2304** shows the response of the GWO: PI-PD controller, and curve **2306** illustrates the MPC-PI controller response. The MPC-(1+PIDN) controller response is presented by curve **2308**. Curve **2308** demonstrates the most efficient sharing of power between the corresponding geographic areas.

[0203] FIG. **24** illustrates a tie-line **23** response between the second geographic area and the third geographic area. Connectivity between the second power system and the third power system is achieved through the tie-line **23**. Various curves shows response by various controllers to sharing of power between two geographic areas through the tie-line. Curve **2402** depicts the GA-PI controller response, curve **2404** shows the response of the GWO: PI-PD controller, and curve **2406** illustrates the MPC-PI controller response. The MPC-(1+PIDN) controller response is presented by curve **2408**. Curve **2408** demonstrates the most efficient sharing of power between the corresponding geographic areas.

[0204] FIG. **25** illustrates an electrical tie-line **14** response between the first geographic area and the fourth geographic area. Connectivity between the first power system and the fourth power system is achieved through the tie-line **14**. Various curves shows response by various controllers to sharing of power between two geographic areas through the tie-line. Curve **2502** depicts the GA-PI controller response, curve **2504** shows the response of the GWO: PI-PD controller, and curve **2506** illustrates the MPC-PI controller response. The MPC-(1+PIDN) controller response is presented by curve **2508**. Curve **2508** demonstrates the most efficient sharing of power between the corresponding geographic areas.

[0205] FIG. **26A**-FIG. **26D** illustrate the collective output power response from various energy sources integrated into the grid.

[0206] In FIG. **26A**, curve **2602** displays the output power of a photovoltaic (PV) source versus time in seconds. Curve **2602** indicates the maximum power attainable at Maximum Power Point Tracking (MPPT) and the corresponding output at 50% irradiation. The time in seconds is plotted on the x-axis, and the power in per unit (p.u.) is plotted on the y-axis, illustrating the variation in power output with changing solar irradiance.

[0207] In FIG. **26B**, curve **2604** shows the wind power generation (in pu.MW) versus time in seconds. The power output is marked by its variability, which is characteristic of wind energy due to fluctuating wind speeds.

[0208] In FIG. **26C**, curve **2606** combines the total output power from thermal energy with PV and wind power contributions versus time in seconds. The curve **2606** illustrates reaction of the overall power output to a change in PV irradiance over the course of time, indicating the combined influence of renewable and conventional energy sources on total power output.

[0209] In FIG. **26D**, curve **2608** presents the total output power when hydropower is included with PV and wind

versus time in seconds. The graph illustrates the variations in power output, particularly highlighting the impact of PV irradiance changes on the combined energy output.

[0210] The present disclosure presents a control approach for a high-order interconnected area (HOIA), where the complexity of a power system was elevated by integrating renewable energy sources (RES) into the power system. Introducing a hydropower plant in the fourth geographic area introduces a right-half-plane zero into the open-loop system, complicating control efforts. To manage such complex network effectively, a robust control system was necessary to counteract frequency deviations resulting from load imbalances. A layered control architecture that combines a MPC with a 1+PIDN controller is demonstrated in the present disclosure. The MPC-(1+PIDN) configuration addresses the demands of the HOIA system, normalizing frequency variations caused by load imbalances in under and about 0.86 seconds, in contrast to the slower responses of other controllers, such as the GWO: PI-PD controller at about 3.3 seconds, the MPC/PI controller at about 9.57 seconds, and the GA-PI controller at about 17.87 seconds. Performance of the controller was evaluated under a variety of load-change scenarios, demonstrating its efficacy when compared to other advanced controllers.

[0211] In a first embodiment, a load frequency control system for a multi-region power system incorporating a plurality of renewable energy sources, comprising a first power system located in a first geographic region (area-1), a second power system located in a second geographic region (area-2), a third power system located a third geographic region (area-3), a fourth power system located in a fourth geographic region (area-4), a first electrical tie-line configured to connect the first power system with the second power system, a second electrical tie-line configured to connect the first power system with the third power system, a third electrical tie-line configured to connect the second power system with the third power system, a fourth electrical tie-line configured to connect the first power system with the fourth power system, where each power system is configured to generate area control signals (ACE), a MPC operatively connected to receive the ACE signals from each electrical tie-line, where the MPC includes a memory storing program instructions, and at least one processor configured to execute the program instructions to generate a control signal $u(k)$, a one plus proportional integral derivative (1+PIDN) controller operatively connected to the MPC, wherein the 1+PIDN controller includes a set of amplifiers having gain parameters, and an optimizer operatively connected to the MPC, wherein the optimizer is configured to receive the ACE signals, apply the ACE signals to a salp swarm algorithm and update the gain parameters of the 1+PIDN controller, where the 1+PIDN controller is configured to receive the control signal $u(k)$ and the updated gain parameters and generate an output signal configured to mitigate power disturbances due to frequency disturbances on the first electrical tie line, the second electrical tie-line, the third electrical tie-line and the fourth electrical tie-line.

[0212] In one aspect, the load frequency control system further comprises a first thermal generator, a first photovoltaic energy generator, a first wind energy generator and a first load located in the first power system, a second thermal generator, a second photovoltaic energy generator, a second wind energy generator and a second load located in the second power system, a third thermal generator, a third photovoltaic energy generator, a third wind energy generator and a third load located in the third power system, and a hydropower plant configured to generate electricity from flowing water, a fourth photovoltaic energy generator, a fourth wind energy generator and a fourth load located in the fourth power system.

[0213] In one aspect, each thermal generator includes a governor, an electrical generator, a steam turbine, and a reheater.

[0214] In one aspect, the controller gain parameters include a gain of one, a proportional gain $K_{sub.p}$, an integral gain $K_{sub.i}$, and a derivative gain $K_{sub.D}$ and a filter factor N .

[0215] In one aspect, the output signal is based on the equation:

$$[00025] \text{output} = (1 + K_P + \frac{K_I}{s} + \frac{K_D N s}{N + s}) \times u(k).$$

[0216] In one aspect, the program instructions include a cost function $J_{sub.ITAE}$, where the optimizer is configured to receive the cost function $J_{sub.ITAE}$ and minimize the cost function $J_{sub.ITAE}$ to update the gain parameters.

[0217] In one aspect, the cost function $J_{sub.ITAE}$ is an integral time absolute error given by:

[00026]

$$J_{ITAE} = \int_0^{T_{sin}} t (\text{.Math. } ACE_1(t) \text{ .Math. } + \text{.Math. } ACE_2(t) \text{ .Math. } + \text{.Math. } ACE_3(t) \text{ .Math. } + \text{.Math. } ACE_4(t) \text{ .Math. }) dt$$

[0218] where t is time, $ACE_{sub.1}$ is the ACE signal generated by the first power system, $ACE_{sub.2}$ is the ACE signal generated by the second power system, $ACE_{sub.3}$ is the ACE signal generated by the third power system, $ACE_{sub.4}$ is the ACE signal generated by the fourth power system.

[0219] In one aspect, each power system includes a busbar and a load connected to the busbar, where the busbar is connected one or more of electrical tie-lines, and where the power disturbances are due to changes in a demand of the load located in the respective power system.

[0220] In one aspect, the first power system comprises a first busbar connected to receive the power disturbances from each of the first tie-line, the second tie-line and the fourth tie-line, a frequency meter configured to measure a frequency on each of the first tie-line, the second tie-line and the fourth tie-line and generate a combined frequency

measurement, a first subtractor configured to subtract the combined frequency measurement from a reference frequency and generate a power to frequency ratio bias signal, an interchange scheduler configured to generate a time window for an exchange of power from the first busbar to the MPC, a second subtractor configured to receive the power disturbances from the first busbar and transmit the power disturbances during the scheduled time window, and a third subtractor configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate the ACE.sub.1 signal.

[0221] In one aspect, the second power system comprises a second busbar connected to receive the power disturbances from each of the first tie-line and the third tie-line, a frequency meter located in the second power system, where the frequency meter is configured to measure a frequency on each of the first tie-line and the third tie-line and generate a combined frequency measurement, a first subtractor configured to subtract the combined frequency measurement from a reference frequency and generate a power to frequency ratio bias signal, an interchange scheduler configured to generate a time window for an exchange of power from the second busbar to the MPC, a second subtractor configured to receive the power disturbances from the second busbar and transmit the power disturbances during the scheduled time window, and a third subtractor configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate the ACE.sub.2 signal.

[0222] In one aspect, the third power system comprises a third busbar connected to receive the power disturbances from each of the second tie-line and the third tie-line, a frequency meter located in the second power system, wherein the frequency meter is configured to measure a frequency on each of the second tie-line and the third tie-line and generate a combined frequency measurement, a first subtractor configured to subtract the combined frequency measurement from a reference frequency and generate a power to frequency ratio bias signal, an interchange scheduler configured to generate a time for an exchange of power from the third busbar to the MPC, a second subtractor configured to receive the power disturbances from the third busbar and transmit the power disturbances at the scheduled time, and a third subtractor configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate the ACE.sub.3 signal.

[0223] In one aspect, the fourth power system comprises a fourth busbar connected to receive the power disturbances from the fourth tie-line, a frequency meter located in the second power system, wherein the frequency meter is configured to measure a frequency on the fourth tie-line and generate a frequency measurement, a first subtractor configured to subtract the frequency measurement from a reference frequency and generate a power to frequency ratio bias signal, an interchange scheduler configured to generate a time window for an exchange of power from the fourth busbar to the MPC, a second subtractor configured to receive the power disturbances from the fourth busbar and transmit the power disturbances during the scheduled time window, and a third subtractor configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate the ACE.sub.3 signal.

[0224] In a second embodiment, a method for load frequency control of a multi-region power system incorporating a plurality of renewable energy sources comprises connecting a first power system located in a first geographic region to a second power system located in a second geographic region by a first electrical tie-line, connecting the first power system to a third power system located in a third geographic region by a second electrical tie-line, connecting the second power system to the third power system by a third electrical tie-line, connecting the first power system to a fourth power system located in a fourth geographic region by a fourth electrical tie-line, generating, by each power system, area control signals (ACE), receiving, by a model predictive controller (MPC) including a memory storing program instructions and at least one processor configured to execute the program instructions, the ACE signals from each power system, generating, by the MPC, a control signal $u(k)$, connecting a one plus proportional integral derivative (1+PIDN) controller which includes a set of amplifiers having gain parameters to the MPC, connecting an optimizer to the MPC receiving, by the optimizer, the ACE signals, applying, by the optimizer, the ACE signals to a salp swarm algorithm, updating, by the optimizer, the gain parameters, receiving, by the 1+PIDN controller, the control signal $u(k)$ and the updated gain parameters, and generating, by the 1+PIDN controller, an output signal configured to mitigate power disturbances due to frequency imbalances on the first electrical tie line, the second electrical tie-line, the third electrical tie-line and the fourth electrical tie-line.

[0225] In one aspect, the method further comprises calculating, by the 1+PIDN controller, the output signal based on the equation:

[00027]
$$\text{output} = (1 + K_P + \frac{K_I}{s} + \frac{K_D N s}{N + s}) \times u(k),$$
 [0226] where K.sub.p is a proportional gain, K.sub.i is an integral gain, K.sub.D is a derivative gain and N is a filter factor.

[0227] In one aspect, the method further comprises fetching, by the optimizer, a cost function J.sub.ITAE from the program instructions, where the cost function J.sub.ITAE is an integral time absolute error given by:

[00028]

$$J_{ITAE} = \int_0^{T_{Sim}} t (\text{Math. ACE}_1(t) \cdot \text{Math.} + \text{Math. ACE}_2(t) \cdot \text{Math.} + \text{Math. ACE}_3(t) \cdot \text{Math.} + \text{Math. ACE}_4(t) \cdot \text{Math.}) dt$$

[0228] where t is time, ACE.sub.1 is the ACE signal generated by the first power system, ACE.sub.2 is the ACE signal generated by the second power system, ACE.sub.3 is the ACE signal generated by the third power system,

ACE.sub.4 is the ACE signal generated by the fourth power system, minimizing, by the optimizer, the cost function J.sub.ITAE, and updating each of the gain parameters based on minimizing the cost function J.sub.ITAE.

[0229] In an aspect, the method further comprising connecting a load to a busbar in each power system, connecting the busbar to one or more of the electrical tie-lines, measuring, by a frequency meter, a frequency imbalance on each of the one or more of the electrical tie-lines due to changes in a demand of the load located in the respective power system, generating, by the frequency meter, a combined frequency imbalance measurement, subtracting, by a first subtractor, the combined frequency imbalance measurement from a reference frequency and generating a power to frequency ratio bias signal, receiving, by the busbar, a combined power disturbance signal on the electrical tie-lines connected to the busbar, generating, by an interchange scheduler, a scheduled time window for an exchange of the combined power disturbance signal to the MPC, receiving, by a second subtractor, the combined power disturbance signal and the scheduled time window, transmitting, by the second subtractor, the combined power disturbance signal at the scheduled time window, receiving, by a third subtractor, the combined power disturbance signal at the scheduled time window and the power to frequency ratio bias signal, subtracting, by a third subtractor, the combined power disturbance signal from the power to frequency ratio bias signal, generating, by the third subtractor, the ACE signal of the respective power system, and transmitting, by the third subtractor, the ACE signal of the respective power system to the MPC.

[0230] In one aspect, the method further comprising connecting, in the first power system, a first thermal generator, a first photovoltaic energy generator, a first wind energy generator and a first load to a first busbar, connecting, in the second power system, a second thermal generator, a second photovoltaic energy generator, a second wind energy generator and a second load to a second busbar, connecting, in the third power system, a third thermal generator, a third photovoltaic energy generator, a third wind energy generator and a third load to a third busbar, and connecting, in the fourth power system, a hydropower plant configured to generate electricity from flowing water, a fourth photovoltaic energy generator, a fourth wind energy generator and a fourth load to a fourth busbar.

[0231] In a third embodiment, a hybrid load frequency controller for interconnected multi-region power systems incorporating a plurality of renewable energy sources, comprising a MPC operatively connected to receive area control error (ACE) signals from electrical tie-lines which interconnect the multi-region power systems, where the MPC includes a memory storing program instructions, and at least one processor configured to execute the program instructions to generate a control signal $u(k)$, a one plus proportional integral derivative (1+PIDN) controller operatively connected to the MPC, where the 1+PIDN controller includes a set of amplifiers having gain parameters, and an optimizer operatively connected to the MPC, where the optimizer is configured to receive the ACE signals, apply the ACE signals to a salp swarm algorithm and update the gain parameters of the 1+PIDN controller, where the 1+PIDN controller is configured to receive the control signal $u(k)$ and the updated gain parameters and generate an output signal configured to mitigate power disturbances due to frequency imbalances on the electrical tie-lines which interconnect the multi-region power systems.

[0232] In one aspect, the hybrid load frequency controller further comprising a cost function J.sub.ITAE stored within the program instructions, where the optimizer is configured to fetch the cost function J.sub.ITAE, apply the cost function J.sub.ITAE to the salp swarm algorithm, and execute the salp swarm algorithm to minimize the cost function J.sub.ITAE.

[0233] In one aspect, the cost function J.sub.ITAE is an integral time absolute error given by:

[00029]

$$J_{ITAE} = \int_0^{T_{Sim}} t(.Math. ACE_1(t) .Math. + .Math. ACE_2(t) .Math. + .Math. ACE_3(t) .Math. + .Math. ACE_4(t) .Math.) dt$$

[0234] where t is time, ACE.sub.1 is an ACE signal generated by a first power system of the interconnected multi-region power systems, ACE.sub.2 is an ACE signal generated by a second power system of the interconnected multi-region power systems, ACE.sub.3 is an ACE signal generated by a third power system of the interconnected multi-region power systems, and ACE.sub.4 is an ACE signal generated by a fourth power system of the interconnected multi-region power systems.

[0235] Next, further details of the hardware description of the computing environment according to exemplary embodiments are described with reference to FIG. 27. In FIG. 27, a controller 2700 is described is representative of the MPC-(1+PIDN) controller 508 of FIG. 5B of the system in which the controller is a computing device which includes a CPU 2701 which performs the processes described above/below. The process data and instructions may be stored in memory 2702. These processes and instructions may also be stored on a storage medium disk 2704 such as a hard drive (HDD) or portable storage medium or may be stored remotely.

[0236] Further, the claims are not limited by the form of the computer-readable media on which the instructions of the inventive process are stored. In an example, the instructions may be stored on CDs, DVDs, in FLASH memory, RAM, ROM, PROM, EPROM, EEPROM, hard disk or any other information processing device with which the computing device communicates, such as a server or computer.

[0237] Further, the claims may be provided as a utility application, background daemon, or component of an operating system, or combination thereof, executing in conjunction with CPU 2701, 2703 and an operating system

such as Microsoft Windows 7, Microsoft Windows 10, Microsoft Windows 11, UNIX, Solaris, LINUX, Apple MAC-OS and other systems known to those skilled in the art.

[0238] The hardware elements in order to achieve the computing device may be realized by various circuitry elements, known to those skilled in the art. In an example, CPU **2701** or CPU **2703** may be a Xenon or Core processor from Intel of America or an Opteron processor from AMD of America, or may be other processor types that would be recognized by one of ordinary skill in the art. Alternatively, the CPU **2701**, **2703** may be implemented on an FPGA, ASIC, PLD or using discrete logic circuits, as one of ordinary skill in the art would recognize. Further, CPU **2701**, **2703** may be implemented as multiple processors cooperatively working in parallel to perform the instructions of the inventive processes described above.

[0239] The computing device in FIG. **27** also includes a network controller **2706**, such as an Intel Ethernet PRO network interface card from Intel Corporation of America, for interfacing with network **2760**. As can be appreciated, the network **2760** can be a public network, such as the Internet, or a private network such as an LAN or WAN network, or any combination thereof and can also include PSTN or ISDN sub-networks. The network **2760** can also be wired, such as an Ethernet network, or can be wireless such as a cellular network including EDGE, 3G, 4G and 5G wireless cellular systems. The wireless network can also be WiFi, Bluetooth, or any other wireless form of communication that is known.

[0240] The computing device further includes a display controller **2708**, such as a NVIDIA GeForce GTX or Quadro graphics adaptor from NVIDIA Corporation of America for interfacing with display **2710**, such as a Hewlett Packard HPL2445w LCD monitor. A general purpose I/O interface **2712** interfaces with a keyboard and/or mouse **2714** as well as a touch screen panel **2716** on or separate from display **2710**. General purpose I/O interface also connects to a variety of peripherals **2718** including printers and scanners, such as an OfficeJet or DeskJet from Hewlett Packard.

[0241] A sound controller **2720** is also provided in the computing device such as Sound Blaster X-Fi Titanium from Creative, to interface with speakers/microphone **2722** thereby providing sounds and/or music.

[0242] The general purpose storage controller **2724** connects the storage medium disk **2704** with communication bus **2726**, which may be an ISA, EISA, VESA, PCI, or similar, for interconnecting all of the components of the computing device. A description of the general features and functionality of the display **2710**, keyboard and/or mouse **2714**, as well as the display controller **2708**, storage controller **2724**, network controller **2706**, sound controller **2720**, and general purpose I/O interface **2712** is omitted herein for brevity as these features are known.

[0243] The exemplary circuit elements described in the context of the present disclosure may be replaced with other elements and structured differently than the examples provided herein. Moreover, circuitry configured to perform features described herein may be implemented in multiple circuit units (e.g., chips), or the features may be combined in circuitry on a single chipset, as shown on FIG. **28**.

[0244] FIG. **28** shows a schematic diagram of a data processing system, according to certain embodiments, for performing the functions of the exemplary embodiments. The data processing system is an example of a computer in which code or instructions implementing the processes of the illustrative embodiments may be located.

[0245] In FIG. **28**, data processing system **2800** employs a hub architecture including a north bridge and memory controller hub (NB/MCH) **2825** and a south bridge and input/output (I/O) controller hub (SB/ICH) **2820**. The central processing unit (CPU) **2830** is connected to NB/MCH **2825**. The NB/MCH **2825** also connects to the memory **2845** via a memory bus, and connects to the graphics processor **2850** via an accelerated graphics port (AGP). The NB/MCH **2825** also connects to the SB/ICH **2820** via an internal bus (e.g., a unified media interface or a direct media interface). The CPU Processing unit **2830** may contain one or more processors and even may be implemented using one or more heterogeneous processor systems.

[0246] In an example, FIG. **29** shows one implementation of CPU **2830**. In one implementation, the instruction register **2938** retrieves instructions from the fast memory **2940**. At least part of these instructions are fetched from the instruction register **2938** by the control logic **2936** and interpreted according to the instruction set architecture of the CPU **2830**. Part of the instructions can also be directed to the register **2932**. In one implementation the instructions are decoded according to a hardwired method, and in another implementation the instructions are decoded according a microprogram that translates instructions into sets of CPU configuration signals that are applied sequentially over multiple clock pulses. After fetching and decoding the instructions, the instructions are executed using the arithmetic logic unit (ALU) **2934** that loads values from the register **2932** and performs logical and mathematical operations on the loaded values according to the instructions. The results from these operations can be feedback into the register and/or stored in the fast memory **2940**. According to certain implementations, the instruction set architecture of the CPU **2830** can use a reduced instruction set architecture, a complex instruction set architecture, a vector processor architecture, a very large instruction word architecture. Furthermore, the CPU **2830** can be based on the Von Neuman model or the Harvard model. The CPU **2830** can be a digital signal processor, an FPGA, an ASIC, a PLA, a PLD, or a CPLD. Further, the CPU **2830** can be an x86 processor by Intel or by AMD; an ARM processor, a Power architecture processor by, e.g., IBM; a SPARC architecture processor by Sun Microsystems or by Oracle; or other known CPU architecture.

[0247] Referring again to FIG. **28**, the data processing system **2800** can include that the SB/ICH **2820** is coupled

through a system bus to an I/O Bus, a read only memory (ROM) **2856**, universal serial bus (USB) port **2864**, a flash binary input/output system (BIOS) **2868**, and a graphics controller **2858**. PCI/PCIe devices can also be coupled to SB/ICH **2888** through a PCI bus **2862**.

[0248] The PCI devices may include, for example, Ethernet adapters, add-in cards, and PC cards for notebook computers. The Hard disk drive **2860** and CD-ROM **2866** can use, for example, an integrated drive electronics (IDE) or serial advanced technology attachment (SATA) interface. In one implementation the I/O bus can include a super I/O (SIO) device.

[0249] Further, the hard disk drive (HDD) **2860** and optical drive **2866** can also be coupled to the SB/ICH **2820** through a system bus. In one implementation, a keyboard **2870**, a mouse **2872**, a parallel port **2878**, and a serial port **2876** can be connected to the system bus through the I/O bus. Other peripherals and devices that can be connected to the SB/ICH **2820** using a mass storage controller such as SATA or PATA, an Ethernet port, an ISA bus, a LPC bridge, SMBus, a DMA controller, and an Audio Codec.

[0250] Moreover, the present disclosure is not limited to the specific circuit elements described herein, nor is the present disclosure limited to the specific sizing and classification of these elements. In an example, the skilled artisan will appreciate that the circuitry described herein may be adapted based on changes on battery sizing and chemistry, or based on the requirements of the intended back-up load to be powered.

[0251] The functions and features described herein may also be executed by various distributed components of a system. In an example, one or more processors may execute these system functions, wherein the processors are distributed across multiple components communicating in a network. The distributed components may include one or more client and server machines, which may share processing, as shown by FIG. **30**, in addition to various human interface and communication devices (e.g., display monitors, smart phones, tablets, personal digital assistants (PDAs)). The network may be a private network, such as a LAN or WAN, or may be a public network, such as the Internet. Input to the system may be received via direct user input and received remotely either in real-time or as a batch process. Additionally, some implementations may be performed on modules or hardware not identical to those described. Accordingly, other implementations are within the scope that may be claimed.

[0252] The above-described hardware description is a non-limiting example of corresponding structure for performing the functionality described herein.

[0253] Numerous modifications and variations of the present disclosure are possible in light of the above teachings. It is therefore to be understood that within the scope of the appended claims, the invention may be practiced otherwise than as specifically described herein.

Claims

1. A load frequency control system for a multi-region power system incorporating a plurality of renewable energy sources, comprising: a first power system located in a first geographic region; a second power system located in a second geographic region; a third power system located in a third geographic region; a fourth power system located in a fourth geographic region; a first electrical tie-line configured to connect the first power system with the second power system; a second electrical tie-line configured to connect the first power system with the third power system; a third electrical tie-line configured to connect the second power system with the third power system; a fourth electrical tie-line configured to connect the first power system with the fourth power system, wherein each power system is configured to generate area control signals (ACE); a model predictive controller (MPC) operatively connected to receive the ACE signals from each electrical tie-line, wherein the MPC includes a memory storing program instructions, and at least one processor configured to execute the program instructions to generate a control signal $u(k)$; a one plus proportional integral derivative (1+PIDN) controller operatively connected to the MPC, wherein the 1+PIDN controller includes a set of amplifiers having gain parameters; and an optimizer operatively connected to the MPC, wherein the optimizer is configured to receive the ACE signals, apply the ACE signals to a salp swarm algorithm and update the gain parameters of the 1+PIDN controller, wherein the 1+PIDN controller is configured to receive the control signal $u(k)$ and the updated gain parameters and generate an output signal configured to mitigate power disturbances due to frequency disturbances on the first electrical tie line, the second electrical tie-line, the third electrical tie-line and the fourth electrical tie-line.
2. The load frequency control system of claim 1, further comprising: a first thermal generator, a first photovoltaic energy generator, a first wind energy generator and a first load located in the first power system; a second thermal generator, a second photovoltaic energy generator, a second wind energy generator and a second load located in the second power system; a third thermal generator, a third photovoltaic energy generator, a third wind energy generator and a third load located in the third power system; and a hydropower plant configured to generate electricity from flowing water, a fourth photovoltaic energy generator, a fourth wind energy generator and a fourth load located in the fourth power system.
3. The load frequency control system of claim 2, wherein each thermal generator includes a governor, an electrical

generator, a steam turbine and a reheater.

4. The load frequency control system of claim 1, wherein the controller gain parameters include a gain of one, a proportional gain $K_{sub.p}$, an integral gain $K_{sub.i}$, and a derivative gain $K_{sub.D}$ and a filter factor N .

5. The load frequency control system of claim 4, wherein the output signal is based on the equation:

$$\text{output} = (1 + K_P + \frac{K_I}{s} + \frac{K_D Ns}{N+s}) \times u(k).$$

6. The load frequency control system of claim 4, wherein the program instructions include a cost function $J_{sub.ITAE}$, wherein the optimizer is configured to receive the cost function $J_{sub.ITAE}$ and minimize the cost function $J_{sub.ITAE}$ to update the gain parameters.

7. The load frequency control system of claim 6, wherein the cost function $J_{sub.ITAE}$ is an integral time absolute error given by:

$$J_{ITAE} = \int_0^{T_{sin}} t(\text{Math. ACE}_1(t) \text{Math.} + \text{Math. ACE}_2(t) \text{Math.} + \text{Math. ACE}_3(t) \text{Math.} + \text{Math. ACE}_4(t) \text{Math.}) dt$$

where t is time, $\text{ACE}_{sub.1}$ is the ACE signal generated by the first power system, $\text{ACE}_{sub.2}$ is the ACE signal generated by the second power system, $\text{ACE}_{sub.3}$ is the ACE signal generated by the third power system, $\text{ACE}_{sub.4}$ is the ACE signal generated by the fourth power system.

8. The load frequency control system of claim 7, wherein each power system includes a busbar and a load connected to the busbar, wherein the busbar is connected one or more of electrical tie-lines, and wherein the power disturbances are due to changes in a demand of the load located in the respective power system.

9. The load frequency control system of claim 8, wherein the first power system comprises: a first busbar connected to receive the power disturbances from each of the first tie-line, the second tie-line and the fourth tie-line; a frequency meter configured to measure a frequency on each of the first tie-line, the second tie-line and the fourth tie-line and generate a combined frequency measurement; a first subtractor configured to subtract the combined frequency measurement from a reference frequency and generate a power to frequency ratio bias signal; an interchange scheduler configured to generate a time window for an exchange of power from the first busbar to the MPC; a second subtractor configured to receive the power disturbances from the first busbar and transmit the power disturbances during the scheduled time window; and a third subtractor configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate the $\text{ACE}_{sub.1}$ signal.

10. The load frequency control system of claim 9, wherein the second power system comprises: a second busbar connected to receive the power disturbances from each of the first tie-line and the third tie-line; a frequency meter located in the second power system, wherein the frequency meter is configured to measure a frequency on each of the first tie-line and the third tie-line and generate a combined frequency measurement; a first subtractor configured to subtract the combined frequency measurement from a reference frequency and generate a power to frequency ratio bias signal; an interchange scheduler configured to generate a time window for an exchange of power from the second busbar to the MPC; a second subtractor configured to receive the power disturbances from the second busbar and transmit the power disturbances during the scheduled time window; and a third subtractor configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate the $\text{ACE}_{sub.2}$ signal.

11. The load frequency control system of claim 10, wherein the third power system comprises: a third busbar connected to receive the power disturbances from each of the second tie-line and the third tie-line; a frequency meter located in the second power system, wherein the frequency meter is configured to measure a frequency on each of the second tie-line and the third tie-line and generate a combined frequency measurement; a first subtractor configured to subtract the combined frequency measurement from a reference frequency and generate a power to frequency ratio bias signal; an interchange scheduler configured to generate a time window for an exchange of power from the third busbar to the MPC; a second subtractor configured to receive the power disturbances from the third busbar and transmit the power disturbances during the scheduled time window; and a third subtractor configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate the $\text{ACE}_{sub.3}$ signal.

12. The load frequency control system of claim 11, wherein the fourth power system comprises: a fourth busbar connected to receive the power disturbances from the fourth tie-line; a frequency meter located in the second power system, wherein the frequency meter is configured to measure a frequency on the fourth tie-line and generate a frequency measurement; a first subtractor configured to subtract the frequency measurement from a reference frequency and generate a power to frequency ratio bias signal; an interchange scheduler configured to generate a time window for an exchange of power from the fourth busbar to the MPC; a second subtractor configured to receive the power disturbances from the fourth busbar and transmit the power disturbances during the scheduled time window; and a third subtractor configured to subtract the power to frequency ratio bias signal from the scheduled power disturbances and generate the $\text{ACE}_{sub.3}$ signal.

13. A method for load frequency control of a multi-region power system incorporating a plurality of renewable energy sources, comprising: connecting a first power system located in a first geographic region to a second power system located in a second geographic region by a first electrical tie-line; connecting the first power system to a third power system located in a third geographic region by a second electrical tie-line; connecting the second power

system to the third power system by a third electrical tie-line; connecting the first power system to a fourth power system located in a fourth geographic region by a fourth electrical tie-line; generating, by each power system, area control signals (ACE); receiving, by a model predictive controller (MPC) including a memory storing program instructions and at least one processor configured to execute the program instructions, the ACE signals from each power system; generating, by the MPC, a control signal $u(k)$; connecting a one plus proportional integral derivative (1+PIDN) controller which includes a set of amplifiers having gain parameters to the MPC; connecting an optimizer to the MPC; receiving, by the optimizer, the ACE signals; applying, by the optimizer, the ACE signals to a salp swarm algorithm; updating, by the optimizer, the gain parameters; receiving, by the 1+PIDN controller, the control signal $u(k)$ and the updated gain parameters; and generating, by the 1+PIDN controller, an output signal configured to mitigate power disturbances due to frequency imbalances on the first electrical tie line, the second electrical tie-line, the third electrical tie-line and the fourth electrical tie-line.

14. The method of claim 13, further comprising: calculating, by the 1+PIDN controller, the output signal based on the equation: $\text{output} = (1 + K_P + \frac{K_I}{s} + \frac{K_D N_s}{N+s}) \times u(k)$, where $K_{\text{sub.p}}$ is a proportional gain, $K_{\text{sub.i}}$ is an integral gain, $K_{\text{sub.p}}$ is a derivative gain and N is a filter factor.

15. The method of claim 14, further comprising: fetching, by the optimizer, a cost function $J_{\text{sub.ITAE}}$ from the program instructions, wherein the cost function $J_{\text{sub.ITAE}}$ is an integral time absolute error given by:

$$J_{\text{ITAE}} = \int_0^{T_{\text{sim}}} t(\text{Math. ACE}_1(t) \text{Math.} + \text{Math. ACE}_2(t) \text{Math.} + \text{Math. ACE}_3(t) \text{Math.} + \text{Math. ACE}_4(t) \text{Math.}) dt$$

where t is time, $\text{ACE}_{\text{sub.1}}$ is the ACE signal generated by the first power system, $\text{ACE}_{\text{sub.2}}$ is the ACE signal generated by the second power system, $\text{ACE}_{\text{sub.3}}$ is the ACE signal generated by the third power system, $\text{ACE}_{\text{sub.4}}$ is the ACE signal generated by the fourth power system; minimizing, by the optimizer, the cost function $J_{\text{sub.ITAE}}$; and updating each of the gain parameters based on minimizing the cost function $J_{\text{sub.ITAE}}$.

16. The method of claim 15, further comprising: connecting a load to a busbar in each power system; connecting the busbar to one or more of the electrical tie-lines; measuring, by a frequency meter, a frequency imbalance on each of the one or more of the electrical tie-lines due to changes in a demand of the load located in the respective power system; generating, by the frequency meter, a combined frequency imbalance measurement; subtracting, by a first subtractor, the combined frequency imbalance measurement from a reference frequency and generating a power to frequency ratio bias signal; receiving, by the busbar, a combined power disturbance signal on the electrical tie-lines connected to the busbar; generating, by an interchange scheduler, a scheduled time window for an exchange of the combined power disturbance signal to the MPC; receiving, by a second subtractor, the combined power disturbance signal and the scheduled time window; transmitting, by the second subtractor, the combined power disturbance signal during the scheduled time window; receiving, by a third subtractor, the combined power disturbance signal during the scheduled time window and the power to frequency ratio bias signal; subtracting, by a third subtractor, the combined power disturbance signal from the power to frequency ratio bias signal; generating, by the third subtractor, the ACE signal of the respective power system; and transmitting, by the third subtractor, the ACE signal of the respective power system to the MPC.

17. The method of claim 13, further comprising: connecting, in the first power system, a first thermal generator, a first photovoltaic energy generator, a first wind energy generator and a first load to a first busbar; connecting, in the second power system, a second thermal generator, a second photovoltaic energy generator, a second wind energy generator and a second load to a second busbar; connecting, in the third power system, a third thermal generator, a third photovoltaic energy generator, a third wind energy generator and a third load to a third busbar; and connecting, in the fourth power system, a hydropower plant configured to generate electricity from flowing water, a fourth photovoltaic energy generator, a fourth wind energy generator and a fourth load to a fourth busbar.

18. A hybrid load frequency controller for interconnected multi-region power systems incorporating a plurality of renewable energy sources, comprising: a model predictive controller (MPC) operatively connected to receive area control error (ACE) signals from electrical tie-lines which interconnect the multi-region power systems, wherein the MPC includes a memory storing program instructions, and at least one processor configured to execute the program instructions to generate a control signal $u(k)$; a one plus proportional integral derivative (1+PIDN) controller operatively connected to the MPC, wherein the 1+PIDN controller includes a set of amplifiers having gain parameters; and an optimizer operatively connected to the MPC, wherein the optimizer is configured to receive the ACE signals, apply the ACE signals to a salp swarm algorithm and update the gain parameters of the 1+PIDN controller, wherein the 1+PIDN controller is configured to receive the control signal $u(k)$ and the updated gain parameters and generate an output signal configured to mitigate power disturbances due to frequency imbalances on the electrical tie-lines which interconnect the multi-region power systems.

19. The hybrid load frequency controller of claim 18, further comprising: a cost function $J_{\text{sub.ITAE}}$ stored within the program instructions, wherein the optimizer is configured to fetch the cost function $J_{\text{sub.ITAE}}$, apply the cost function $J_{\text{sub.ITAE}}$ to the salp swarm algorithm, and execute the salp swarm algorithm to minimize the cost function $J_{\text{sub.ITAE}}$.

20. The hybrid load frequency controller of claim 19, wherein: the cost function $J_{\text{sub.ITAE}}$ is an integral time

absolute error given by:

$$J_{ITAE} = \int_0^{T_{Sim}} t(\text{.Math. ACE}_1(t) \text{.Math.} + \text{.Math. ACE}_2(t) \text{.Math.} + \text{.Math. ACE}_3(t) \text{.Math.} + \text{.Math. ACE}_4(t) \text{.Math.}) dt$$

where t is time, ACE.sub.1 is an ACE signal generated by a first power system of the interconnected multi-region power systems, ACE.sub.2 is an ACE signal generated by a second power system of the interconnected multi-region power systems, ACE.sub.3 is an ACE signal generated by a third power system of the interconnected multi-region power systems, and ACE.sub.4 is an ACE signal generated by a fourth power system of the interconnected multi-region power systems.
